MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

SHIFT REGISTER, DIGITAL, KEY CODE KDP

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic shift register module.

1.2 Classification. This specification covers one class of modules.

Class 1 as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3 REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 1.01 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be KDP.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/59-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc ±1 percent.

3.10 Module speed. When driving the maximum allowable load, the module shall operate at a clock rate from dc to 10 MHz. The high level and low level output voltages shall conform to the limits shown in table I.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

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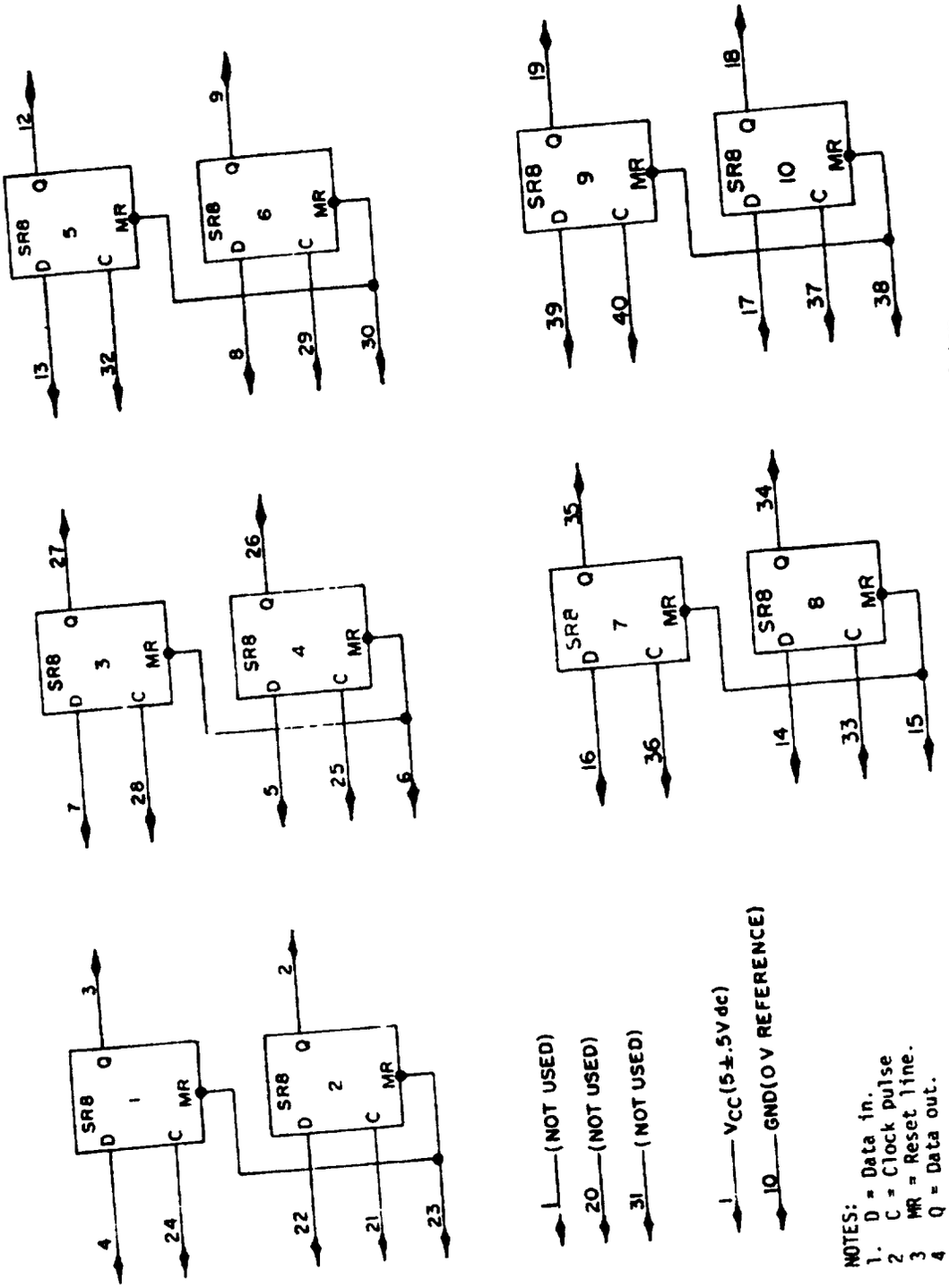


FIGURE 1. KDP module function diagram, shift register.

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* TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A10	V_{OUT} (low loaded)	III	S	0.01	0.80	0.01	0.80	0.01	0.80	0.01	0.80	V
B1-B25	Input breakdown current	IV	S	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	mA
C1-C25	Input clamp diode voltage	V	S	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
D1-D10	Static output voltage, high level	VI	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
E1-E10	Static output voltage, low level	VI	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
F1-F10	Static output voltage, reset	VI	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
G1-G15	Input current, low level	VII	X	0.4	1.73	0.4	1.73	0.4	1.73	0.4	1.73	mA
G16-G25	Input current, low level	VII	X	0.6	2.60	0.6	2.60	0.6	2.60	0.6	2.60	mA
H1-H15	Input current, high level	VIII	X	0	40	0	40	0	40	0	40	µA
H16-H25	Input current, high level	VIII	X	0	60	0	60	0	60	0	60	µA
J1-J10	Output short circuit current	IX	X	12	90	12	90	12	90	12	90	mA
K	Power supply current	X	S	1	410	1	410	1	410	1	410	mA
L1-L10	Transition time (t_{TLH})	XI	S	4	45	4	45	4	50	4	50	ns
M1-M10	Transition time (t_{THL})	XI	S	4	20	4	20	4	25	4	25	ns
N1-N10	Propagation delay time (t_{pLH})	XI	S	4	30	4	30	4	35	4	35	ns
P1-P10	Propagation delay time (t_{pHL})	XI	S	4	46	4	46	4	50	4	50	ns
R1-R10	Setup time	XII	S									
	TA			487	637	487	637	487	637	487	637	ns
	TB			432	590	432	590	432	590	432	590	ns
	TC			900	1100	900	1100	900	1100	900	1100	ns

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
S1-S10	Hold time	XII	S									
	TA			0	50	0	50	0	50	0	50	ns
	TB			432	590	432	590	432	590	432	590	ns
	TC			900	1100	900	1100	900	1100	900	1100	ns
T1-T10	Maximum shift frequency	XIII	S									
	TA			142	205	142	205	142	205	142	205	ns
	TB			80	135	80	135	80	135	80	135	ns
	TC			180	204	180	204	180	204	180	204	ns
U1-U10	Propagation delay time (t_{PHL}), (minimum MR pulsewidth)	XIV	S	4	75	4	75	4	75	4	75	ns
V1-V10	Minimum MR pulsewidth (clock low)	XV	S	7.2	8.8	7.2	8.8	7.2	8.8	7.2	8.8	µs
W1-W10	Output crosstalk, high level	XVI	Q	0	400	0	400	0	400	0	400	mV
X1-X10	Output crosstalk, low level	XVII	Q	0	500	0	500	0	500	0	500	mV
Y	Power supply transient amplitude	XVII	Q	0	600	0	600	0	600	0	600	mV
Z	Power supply transient rise time	XIX	S	1.0	2.0	1.0	2.0	1.0	2.0	1.0	2.0	ms
AA1	Isolation	XX	S	10	∞	1.0	∞	10	∞	1.0	∞	MΩ
AA2-AA4	Isolation	XX	Q	10	∞	1.0	∞	10	∞	1.0	∞	MΩ
BB1-BB25	Input capacitance	XXI	Q	2	15	2	15	2	15	2	15	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through BB shall be performed in any order with the following exceptions. If test conditions A, B, or AA are part of a sequence of tests, the order of testing shall be as follows. A, B, AA where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through BB in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions C through Y and Z shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition Z, shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests C through Z and BB shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests C through BB shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

* 6.3 Cross reference of classification. Military specification MIL-M-28787/59 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing No. 10001-2658774. Part number M28787/59-1 supersedes part numbers 2658774 and M28787/59.

* 6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Maximum power dissipation	2.26 W (outputs not loaded)
Speed	DC to 10 MHz (see 3.10)
Input threshold voltage	
Low level	0.8 Vdc maximum
High level	2.0 Vdc maximum

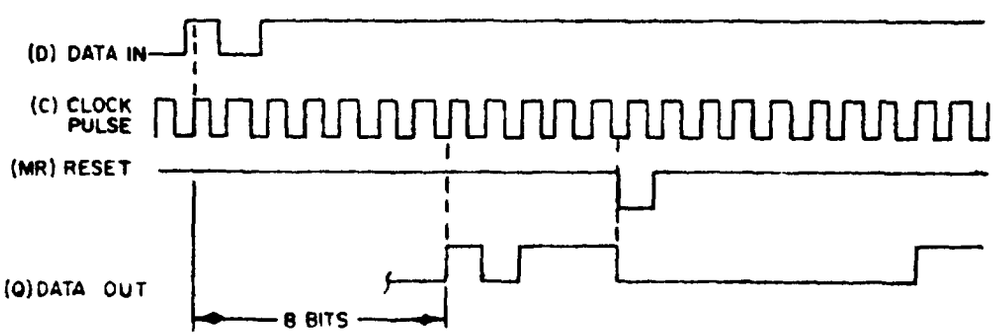
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Input current, low level (V_{IL} = 0 Vdc)	
Data inputs	-1.73 mA maximum
MR inputs	-1.73 mA maximum
Clock inputs	-2.60 mA maximum
Input current, high level (V_{IH} = 2.4 Vdc)	
Data inputs	40 µA maximum
MR inputs	40 µA maximum
Clock inputs	60 µA maximum
Output voltage	
Low level	0 Vdc minimum, 0.4 Vdc maximum
High level	2.4 Vdc minimum, 4.5 Vdc maximum
Output drive	
Low level (V_{OL} at 0.4 Vdc)	9.6 mA maximum
High level (V_{OH} at 2.4 Vdc)	-0.6 mA maximum
Propagation delay time	
t_{PHL} (Clock-to-output)	50 ns maximum
t_{PLH} (Clock-to-output)	35 ns maximum
MR pulsewidth	35 ns maximum
Clock pulsewidth	45 ns maximum

* 6.5 Module function. This module contains ten 8-bit shift registers that have independent clock and data inputs with a common reset for each pair of registers.

6.5.1 Input to output relationship. Each shift register has the input to output relationship of table II for positive logic.

TABLE II. Input to output relationship. 1/ - 3/



Truth table		
t		t+8
D	MR	Q
0	1	0
1	1	1
0	0	0
1	0	0

1/ t = bit time before clock low-to-high transition
2/ t+8 = bit time after 8 clock pulses
3/ MR is asynchronous.

6.5.1.1. Operational description. Clock pulse (C), data in (D), and data out (Q) for each of the 10 registers are brought out to the module pins. One reset (MR) line is common to each pair of registers. The registers trigger on the positive transition of the clock pulse. A low level on the reset line resets all stages of the register such that the output is at a low level.

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6.5.1.2 Setup time (t_{SETUP}) and hold time (t_{HOLD}). The data logic level must be present at least 35 ns prior to the clock low-to-high transition, but it is not necessary to maintain this level after the transition.

6.5.1.3 Recovery time master reset to clock. The reset (MR) input must be high at least 40 ns prior to the clock low-to-high transition.

6.5.2 Parallel outputs. Paralleling of outputs is not permissible.

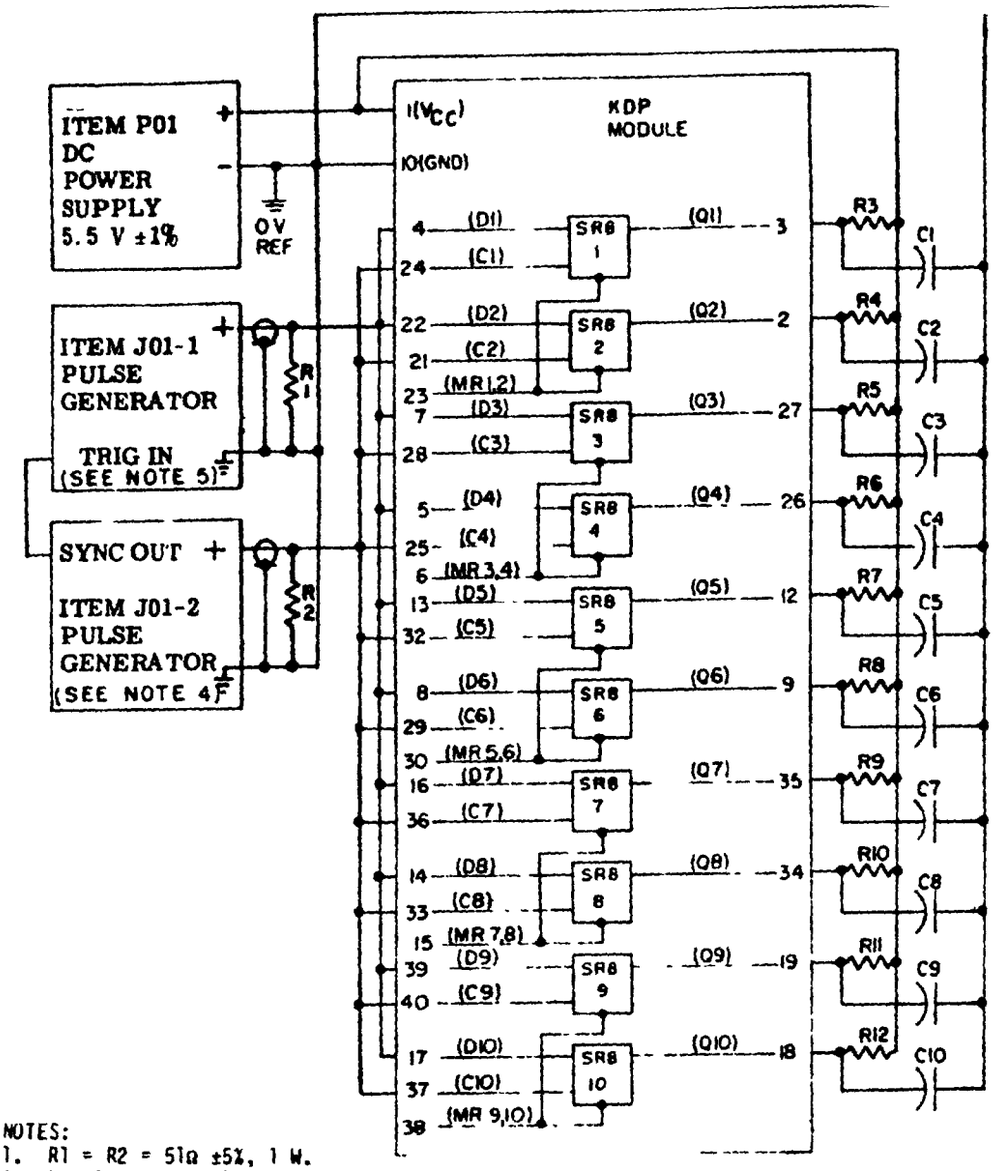
6.3 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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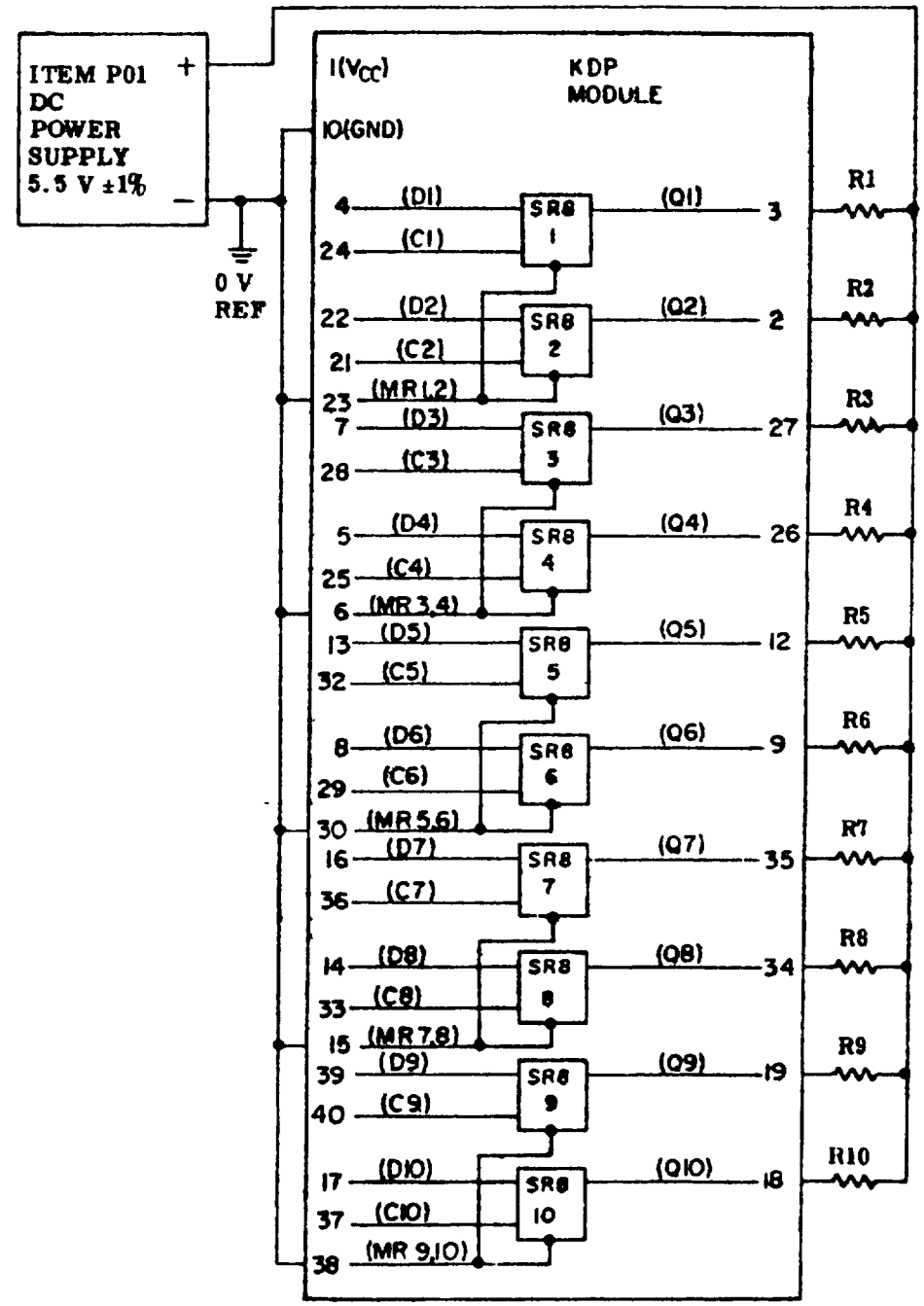
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1 W.
2. R3-R12 = 536Ω ±1%, 1/4 W.
3. C1-C10 = 100 pF ±20%.
4. Clock input pulse parameters.
   a. PA = 3.5 ±0.1 V.
   b. $t_{TLH} = t_{THL}$ = 15 ±5 ns.
   c. PW = 100 ns ±10%.
   d. PRR = 2 MHz ±10%.
5. Data input pulse parameters:
   a. PA = 3.5 ±0.1 V.
   b. $t_{TLH} = t_{THL}$ = 15 ±5 ns.
   c. PW = 500 ns ±10 .
   d. PRR = 1 MHz ±10 .
6. Adjust the time relationship between the input pulses such that the clock rise occurs 200 ns after the data rise (refer to figure 26).

* FIGURE 2. Operating temperature and life test circuit.

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NOTE: R1-R10 = 536Ω ±1%, 1/4 W.

FIGURE 3. Component temperature test circuit.

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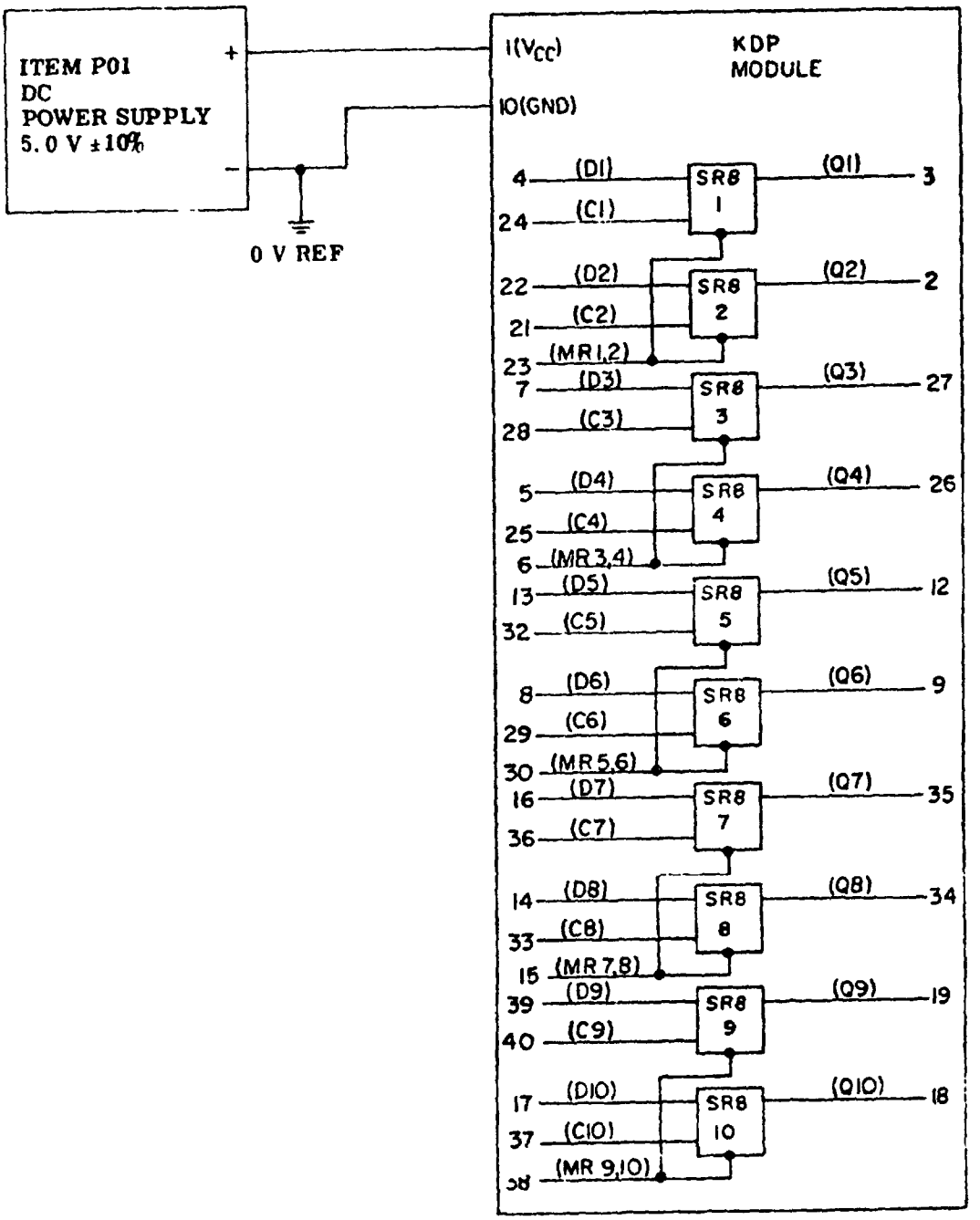


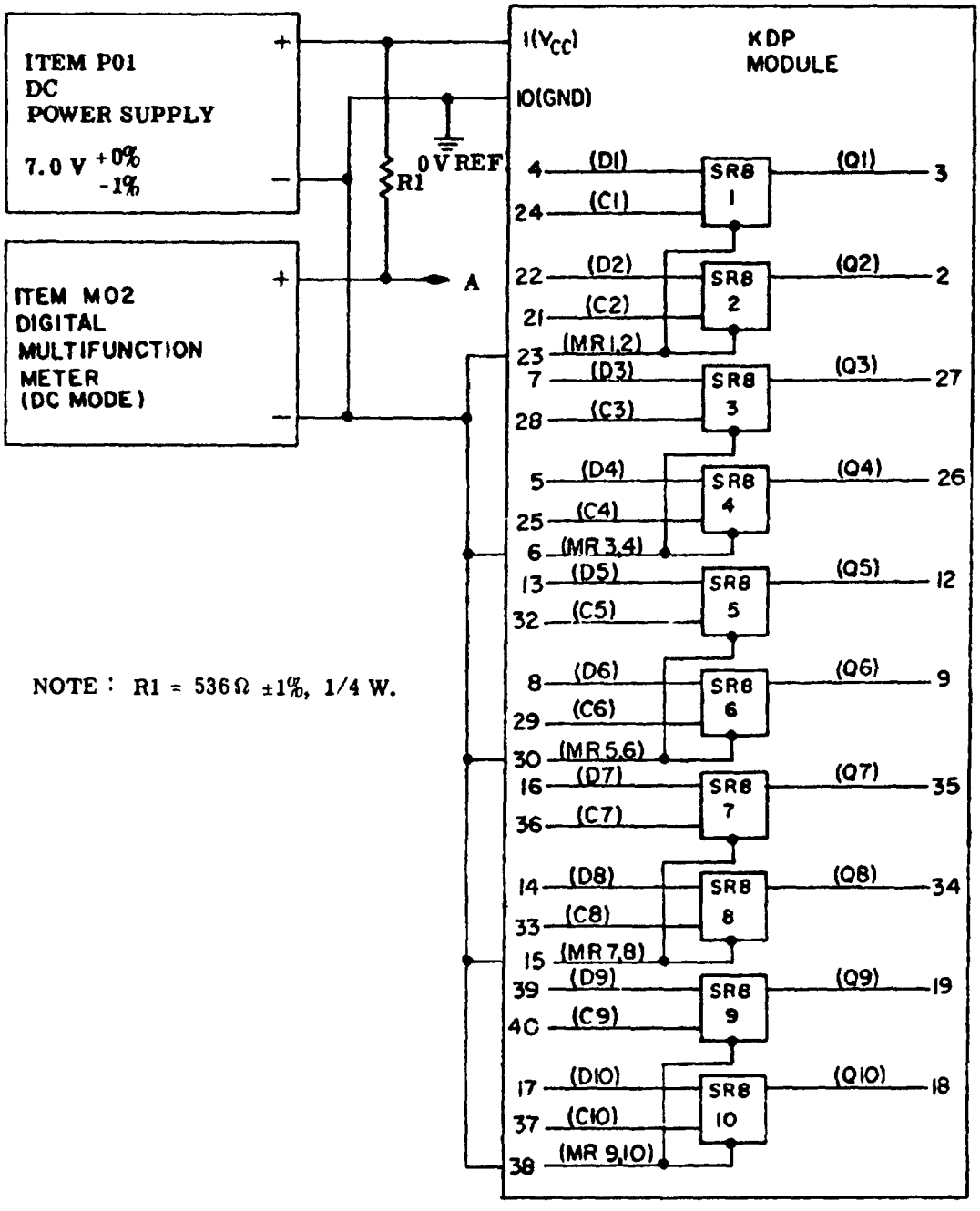
FIGURE 4. Vibration test circuit.

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TABLE III. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
A1	1Q V_{OUT} (low loaded)	3	0.01	0.80	V	5
A2	2Q	2	↓	↓	↓	↓
A3	3Q	27				
A4	4Q	26				
A5	5Q	12				
A6	6Q	9				
A7	7Q	35				
A8	8Q	34				
A9	9Q	19				
A10	10Q	18				

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NOTE : R1 = 536Ω ±1%, 1/4 W.

FIGURE 5. V_{OUT} (low-loaded) test circuit for table III.

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TABLE IV. Input breakdown current test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	1D Input breakdown current	4	21,22,23,24	0.0	1.0	mA	6
B2	1C	24	4,21,22,23				
B3	2D	22	4,21,23,24				
B4	2C	21	4,22,23,24				
B5	2MR	23	4,21,22,24				
B6	3D	7	5,6,25,28				
B7	3C	28	5,6,7,25				
B8	4D	5	6,7,25,28				
B9	4C	25	5,6,7,28				
B10	4MR	6	5,7,25,28				
B11	5D	13	8,29,30,32				
B12	5C	32	8,13,29,30				
B13	6D	8	13,29,30,32				
B14	6C	29	8,13,30,32				
B15	6MR	30	8,13,29,32				
B16	7D	16	14,15,33,36				
B17	7C	36	14,15,16,33				
B18	8D	14	15,16,33,16				
B19	8C	33	14,15,16,36				
B20	8MR	15	14,16,33,36				
B21	9D	39	17,37,38,40				
B22	9C	40	17,37,38,39				

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TABLE IV. Input breakdown current test - Continued.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B23	10D Input breakdown current	17	37,38,39,40	0.0	1.0	mA	6
B24	10C Input breakdown current	37	17,38,39,40	0.0	1.0	mA	6
B25	10MR Input breakdown current	38	17,37,39,40	0.0	1.0	mA	6

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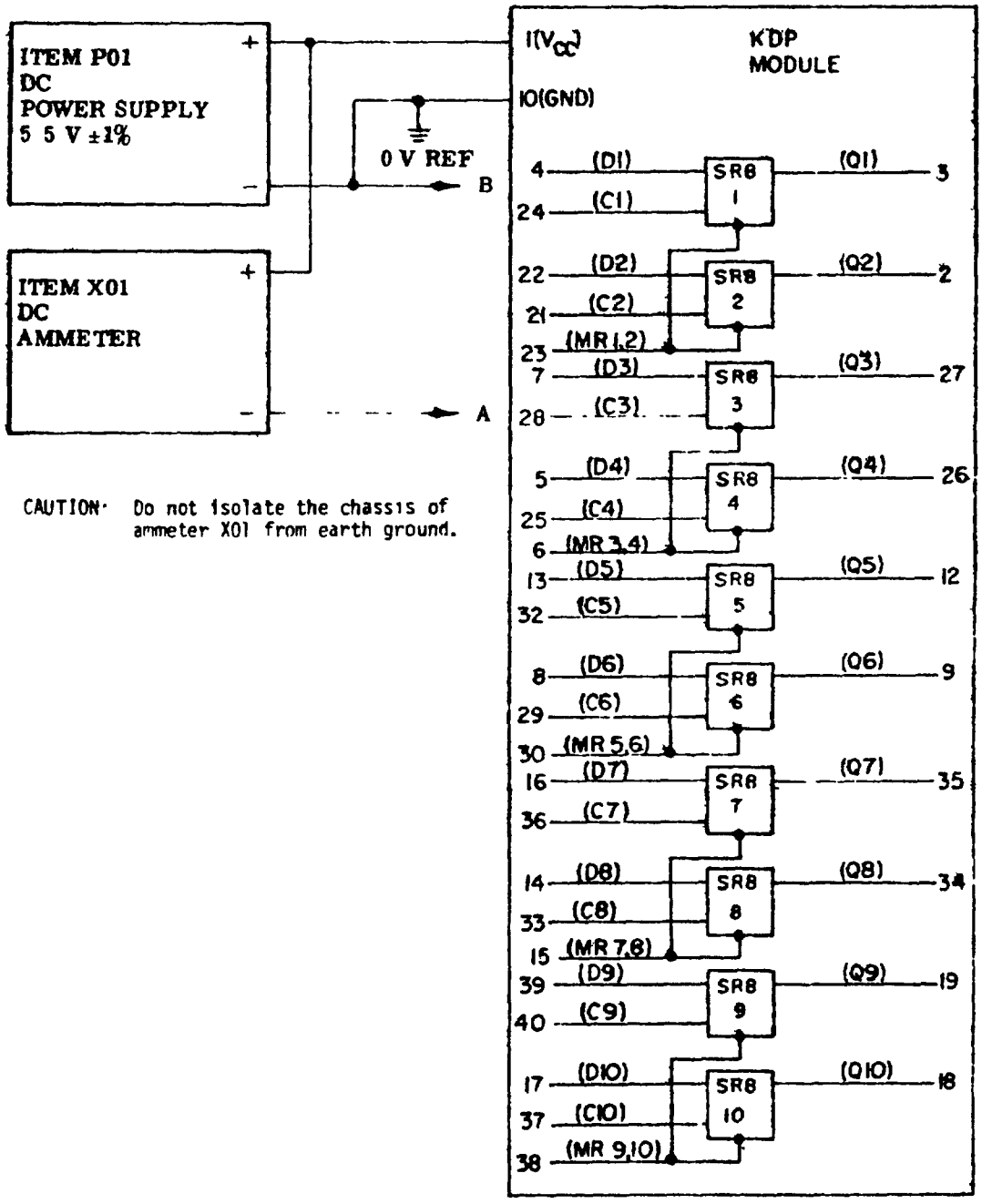


FIGURE 6. Input breakdown current test circuit for table IV.

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TABLE V. Input clamp diode voltage test.

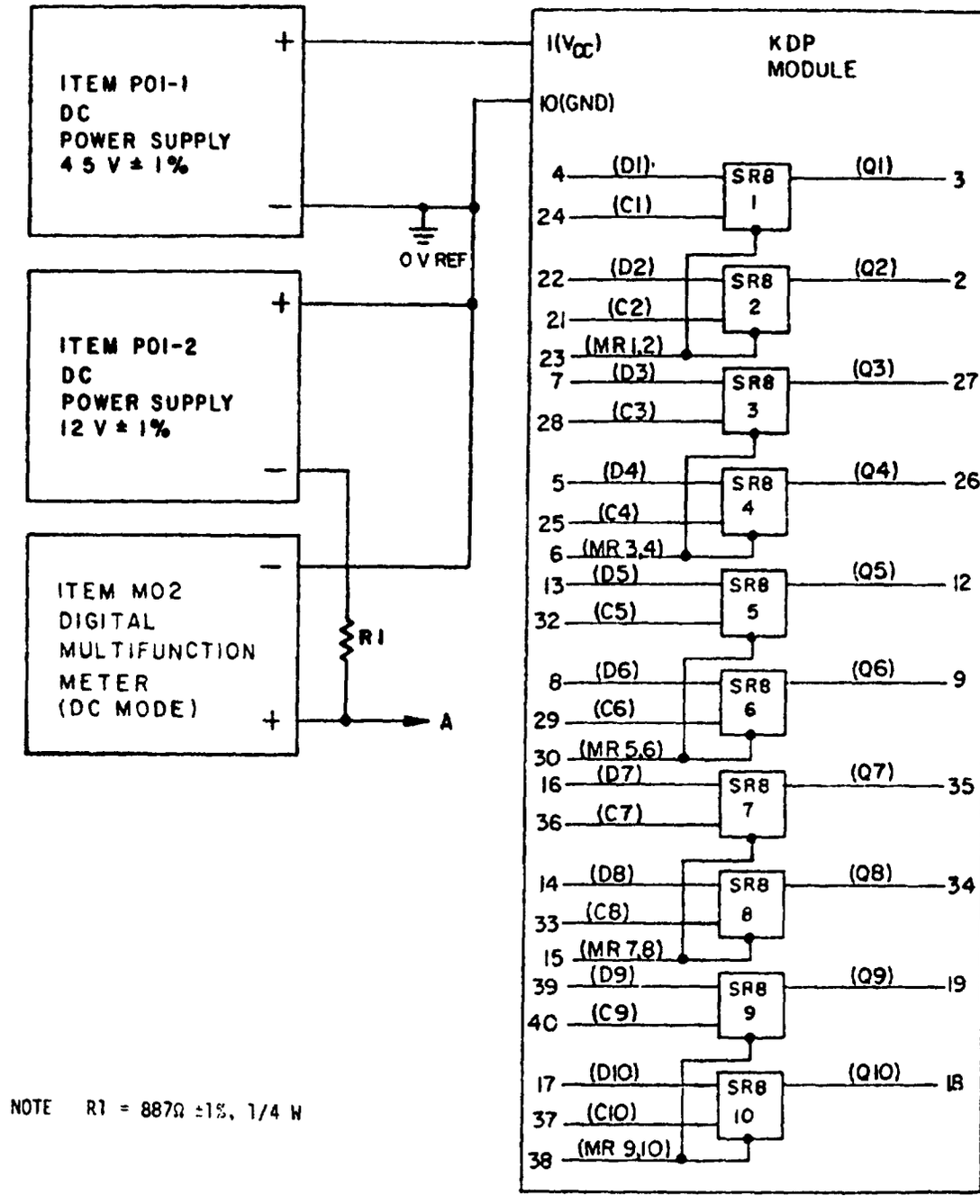
Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
C1	1D Input clamp diode voltage	4	-0.4	-1.5	V	7
C2	1C	24				
C3	2D	22				
C4	2C	21				
C5	1, 2MR	23				
C6	3D	7				
C7	3C	28				
C8	4D	5				
C9	4C	25				
C10	3, 4MR	6				
C11	5D	13				
C12	5C	32				
C13	6D	8				
C14	6C	29				
C15	5, 6MR	30				
C16	7D	16				
C17	7C	36				
C18	8D	14				
C19	8C	33				
C20	7, 8MR	15				
C21	9D	39				
C22	9C	40				

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TABLE V. Input clamp diode voltage test - Continued.

Test condition	Parameter		Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
C23	10D	Input clamp diode voltage	17	-0.4	-1.5	V	7
C24	10C	Input clamp diode voltage	37	-0.4	-1.5	V	7
C25	9, 10MR	Input clamp diode voltage	38	-0.4	-1.5	V	7

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NOTE R1 = 887Ω ±1%, 1/4 W

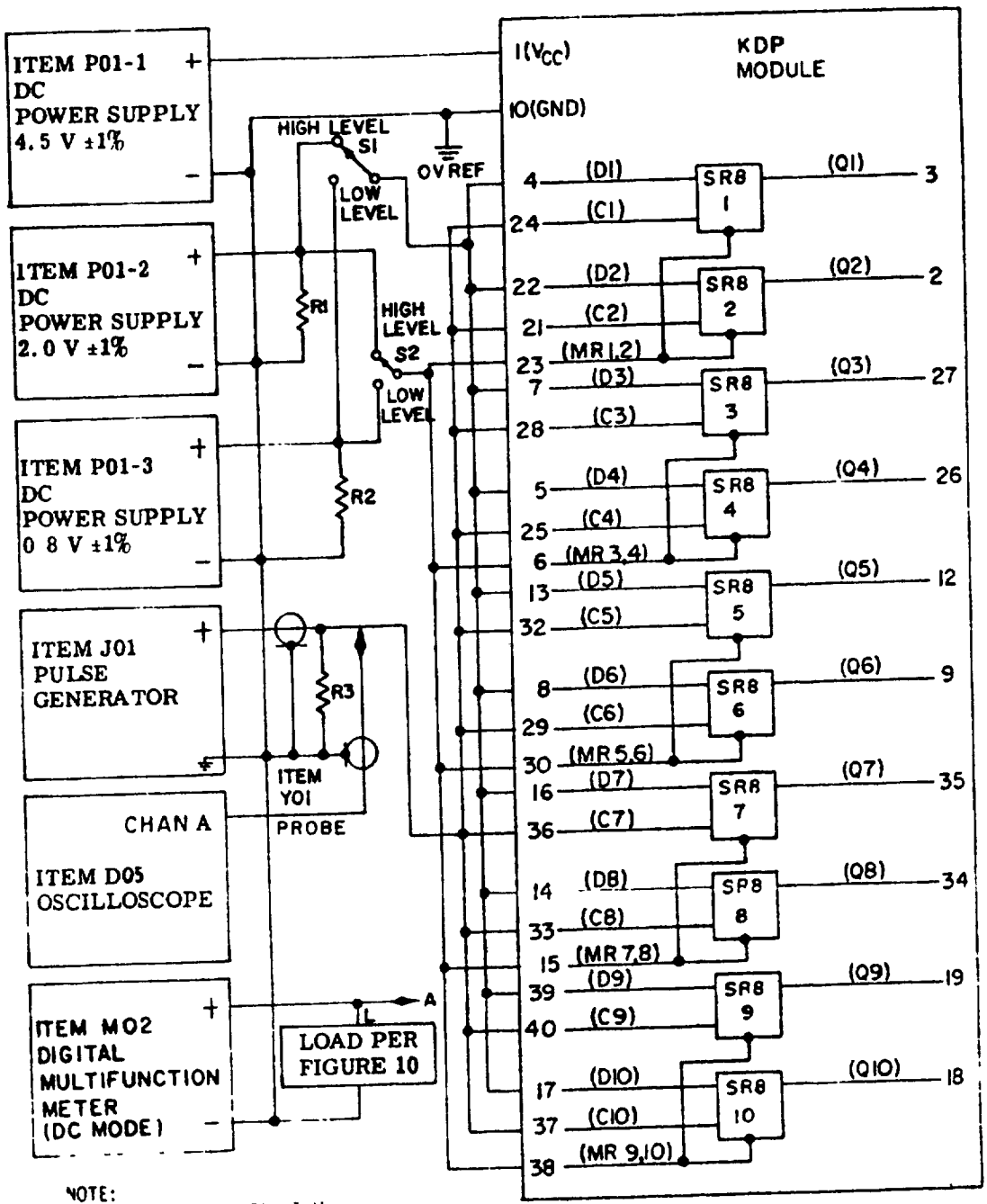
* FIGURE 7. Input clamp diode voltage test circuit for table V.

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* TABLE VI. Static output voltage (high-level, low-level, and reset) test.

Test condition	Parameter		Connect	Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	Min	Max		
D1	1Q	Static output voltage, high level	3	2.4	4.5	V	8, 9
D2	2Q		2				
D3	3Q		27				
D4	4Q		26				
D5	5Q		12				
D6	6Q		9				
D7	7Q		35				
D8	8Q		34				
D9	9Q		19				
D10	10Q		18				
E1	1Q	Static output voltage, low level	3	0.0	0.4		
E2	2Q		2				
E3	3Q		27				
E4	4Q		26				
E5	5Q		12				
E6	6Q		9				
E7	7Q		35				
E8	8Q		34				
E9	9Q		19				
E10	10Q		18				
F1	1Q	Static output voltage, reset	3				
F2	2Q		2				
F3	3Q		27				
F4	4Q		26				
F5	5Q		12				
F6	6Q		9				
F7	7Q		35				
F8	8Q		34				

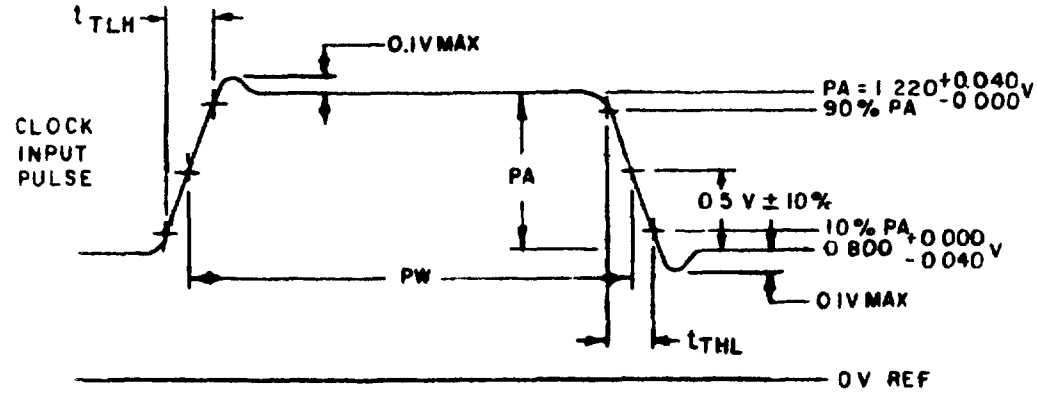
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NOTE:
1. R1-R3 = 51Ω ±5%, 1 W.
2. S1-S2 = SPDT switch.

* FIGURE 8. Static output voltage (high-level, low-level, and reset) test circuit for table VI.

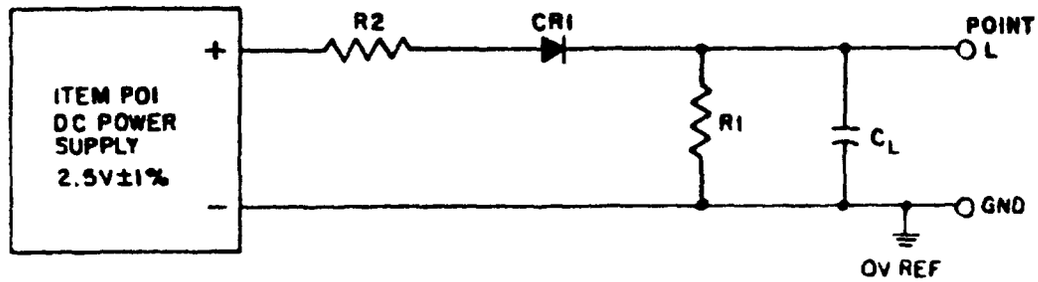
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NOTES:
1. Adjust item J01 output for the indicated clock pulse with the following parameters
   PW = 500 ns ±10%.
   $t_{TLH} = t_{THL}$ = 25 ns ±10%.
   PRR = 30 kHz ±50%.
2. Put item J01 in single pulse mode. Using item M02, adjust the low level.
3. S1 and S2 shall be in high level position initially.
4. Throw S2 to low level position for 100 ns minimum and then back to high level position. Observe that all outputs as indicated on item M02 are at a low level (0.0 to 0.4 V).
5. Apply one clock pulse to the module. Observe that all outputs as indicated on item M02 are at a low level.
6. Throw S1 to low level position.
7. Apply one clock pulse to the module. Observe that all outputs as indicated on item M02 are at a low level.
8. Throw S1 to high level position.
9. Apply six clock pulses to the module. Observe that all outputs as indicated on item M02 are at a low level after each of the first five clock pulses.
10. Connect point A to the module pins as specified in table VI (high level), test condition D.
11. Apply one clock pulse to the module.
12. Connect point A to the module pins as specified in table VI (low level), test condition E.
13. Apply one clock pulse to the module.
14. Observe that all outputs as indicated on item M02 are at a high level (2.4 to 4.5 V).
15. Throw S2 to low level position for a minimum of 100 ns and then back to high level position.
16. Connect point A to the module pins as specified in table VI (reset), test condition F.

* FIGURE 9. Static output voltage (high-level, low-level, and reset) test circuit waveform for table VI.

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NOTES:
1. CR1 = 1N4150, or equivalent.
2. R1 = 4.02 kΩ ±1%, 1/4 W.
3. R2 = select value for current flow of 10.3 mA ±1% form point "L" to ground.
4. C_L = 50 ±5 pF (including probe and parasitic capacitance) for test conditions W and X.
5. C_L = 100 ±5 pF (including probe and parasitic capacitance) for test conditions L, M, N, P, T, U, and V.
6. C_L is not required for test conditions D, E, and F.
7. The maximum lead length from point "L" to the appropriate module pin shall be 3 inche (76.2 mm).
8. During output switching item POI shall have a maximum overshoot of 500 mV.

* FIGURE 10. KDP logic load circuit.

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TABLE VII. Input current (low-level) test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
G1	1D Input current, low level	4	0.4	1.73	mA	11
G2	2D	22				
G3	1, 2MR	23				
G4	3D	7				
G5	4D	5				
G6	3, 4MR	6				
G7	5D	13				
G8	6D	8				
G9	5, 6MR	30				
G10	7D	16				
G11	8D	14				
G12	7, 8MR	15				
G13	9D	39				
G14	10D	17				
G15	9, 10MR	38				
G16	1C	24	0.6	2.60		
G17	2C	21				
G18	3C	28				
G19	4C	25				
G20	5C	32				
G21	6C	29				
G22	7C	36				

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TABLE VII Input current (low-level) test - Continued.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
C23	8C Input current, low level	33	-0.6	2.60	mA	11
C24	9C Input current, low level	40	-0.6	2.60	mA	11
C25	10C Input current, low level	37	-0.6	2.60	mA	11

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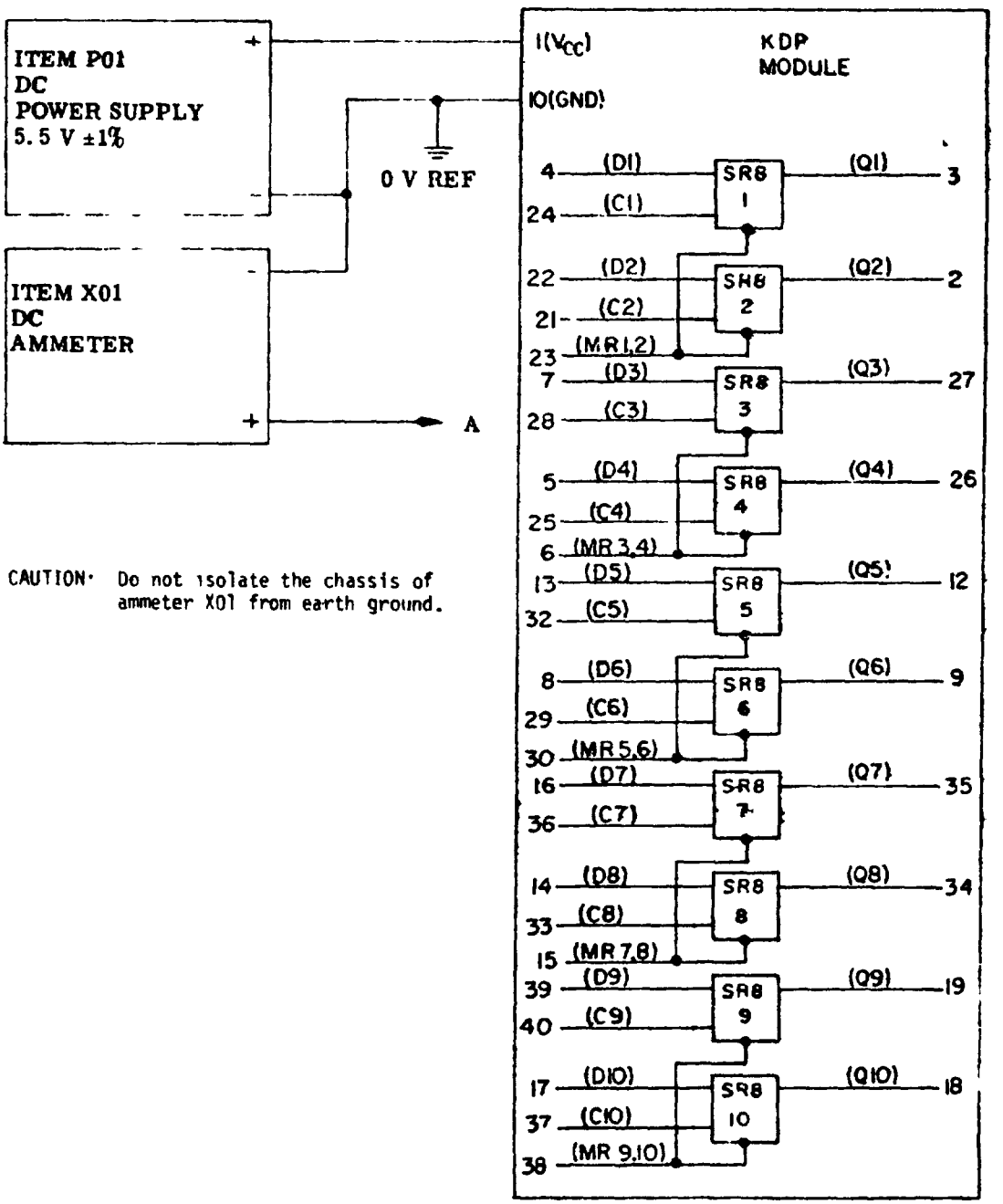


FIGURE 11. Input current, (low-level) test circuit for table VII.

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TABLE VIII. Input current (high-level) test.

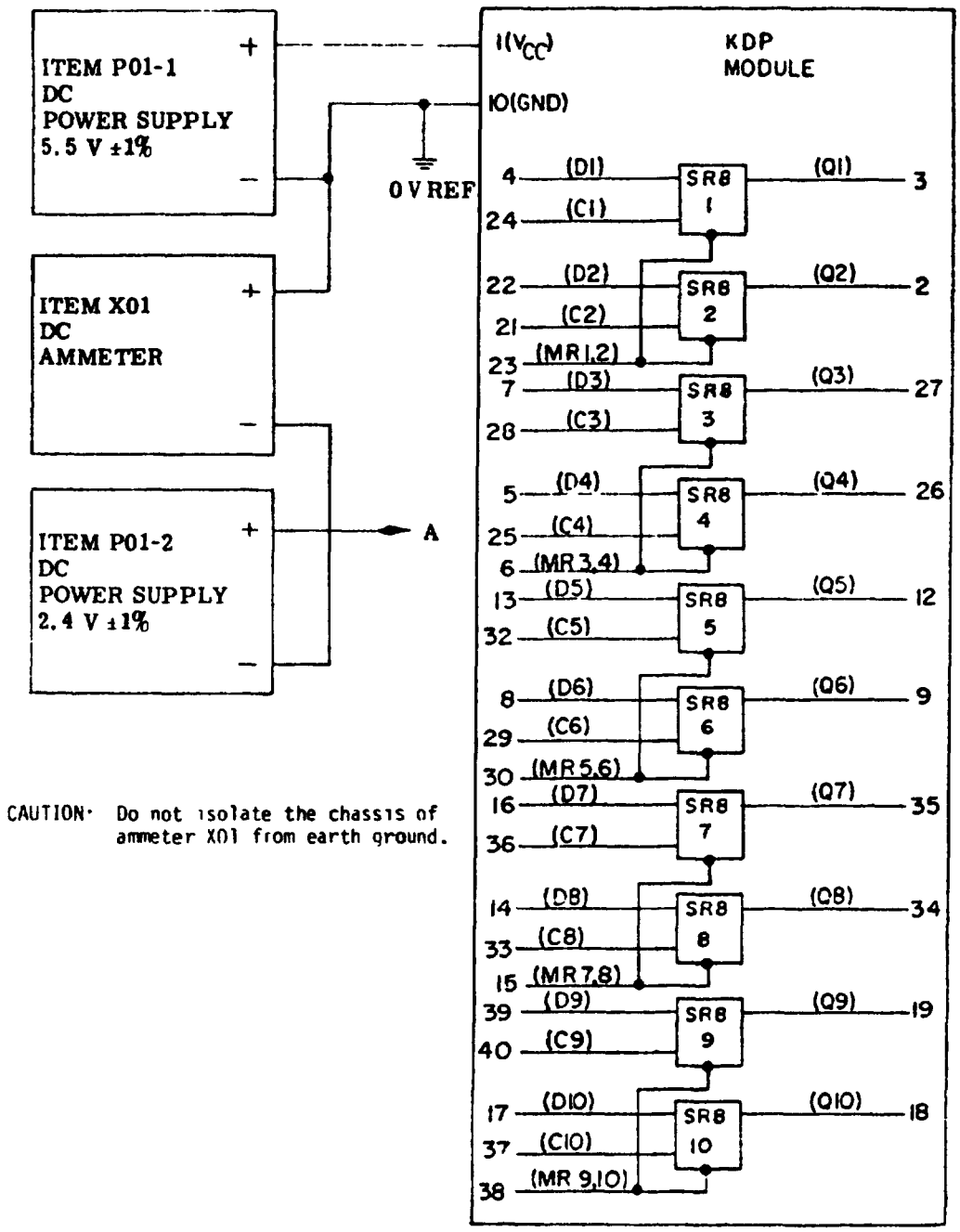
Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
H1	1D Input current, high level	4	0	40	µA	12
H2	2D	22				
H3	1, 2MR	23				
H4	3D	7				
H5	4D	5				
H6	3, 4MR	6				
H7	5D	13				
H8	6D	8				
H9	5, 6MR	30				
H10	7D	16				
H11	8D	14				
H12	7, 8MR	15				
H13	9D	39				
H14	10D	17				
H15	9, 10MR	38				
H16	1C	24		60		
H17	2C	21				
H18	3C	28				
H19	4C	25				
H20	5C	32				
H21	6C	29				
H22	7C	36				

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TABLE VIII. Input current (high-level) test - Continued.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
H23	8C Input current, high level	33	0	60	µA	12
H24	9C Input current, high level	40	0	60	µA	12
H25	10C Input current, high level	37	0	60	µA	12

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CAUTION: Do not isolate the chassis of ammeter X01 from earth ground.

FIGURE 12. Input current (high-level) test circuit for table VIII.

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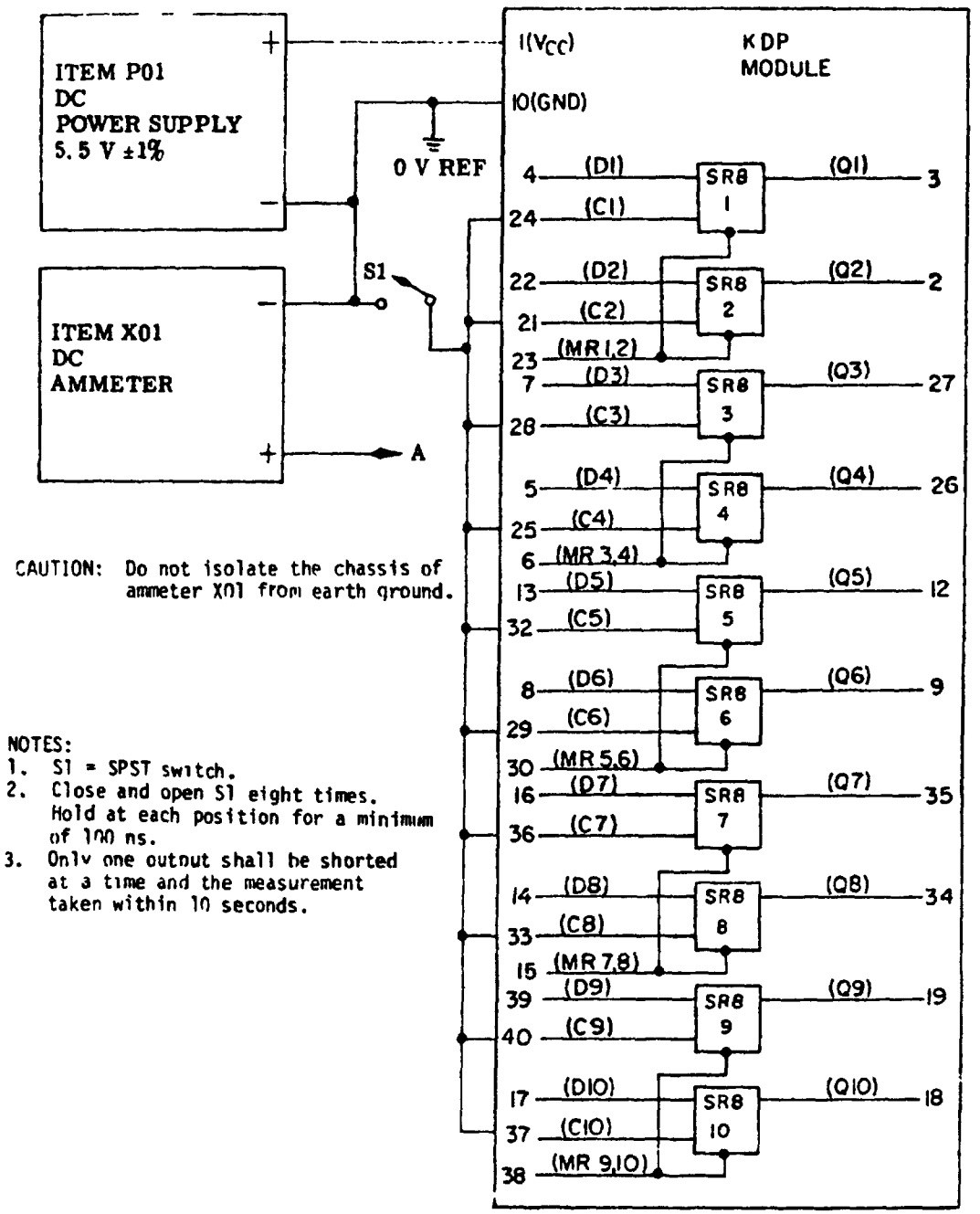
TABLE IX. Output short circuit current test.

Test condition	Parameter		Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
J1	1Q	Output short circuit current	3	12	90	mA	13
J2	2Q		2				
J3	3Q		27				
J4	4Q		26				
J5	5Q		12				
J6	6Q		9				
J7	7Q		35				
J8	8Q		34				
J9	9Q		19				
J10	10Q		18				

TABLE X. Power supply current test.

Test condition	Parameter	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
K	Power supply current	1	410	mA	14

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* FIGURE 13. Output short circuit current test circuit for table IX.

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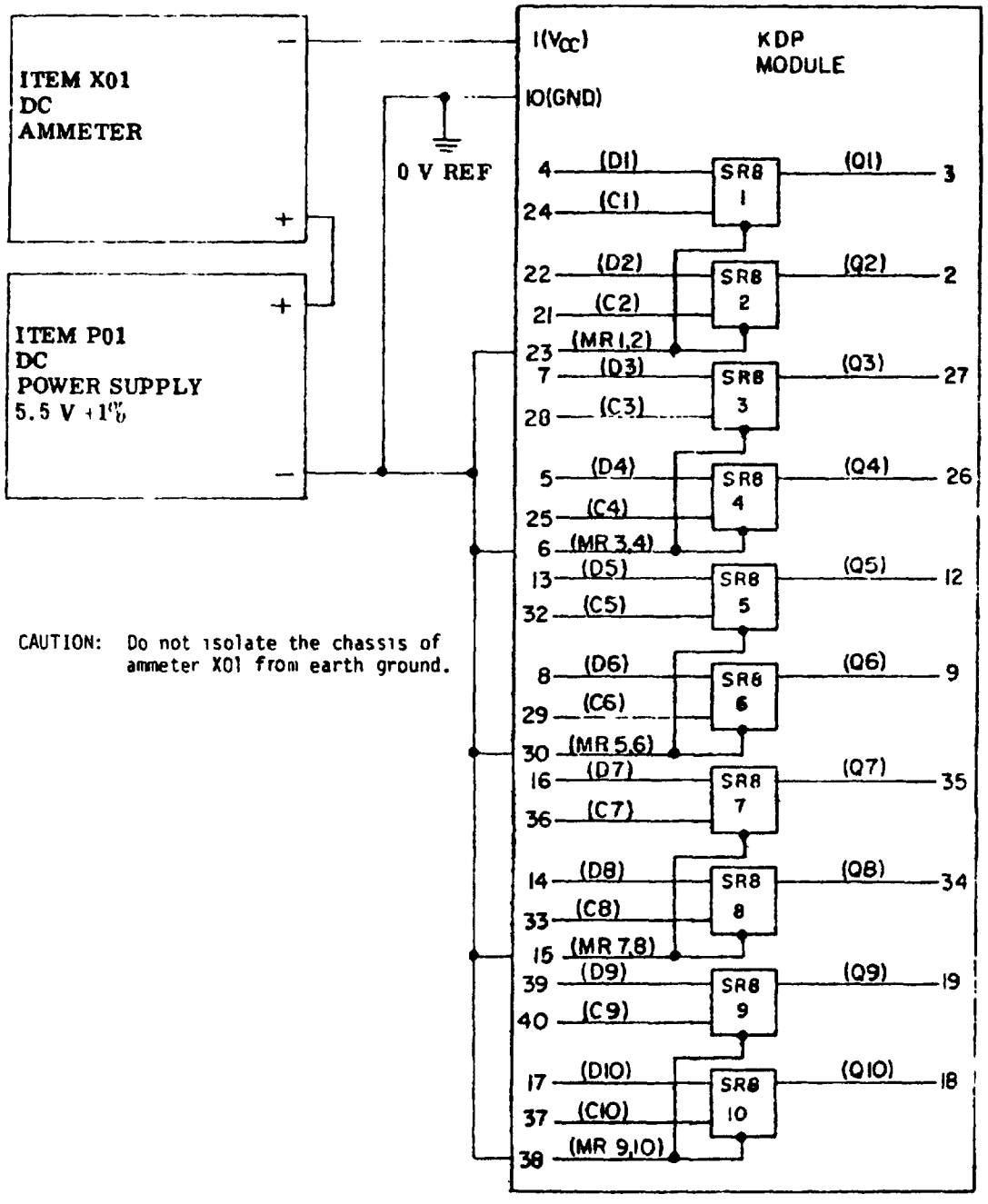


FIGURE 14. Power supply current test circuit for table X.

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TABLE XI. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test.

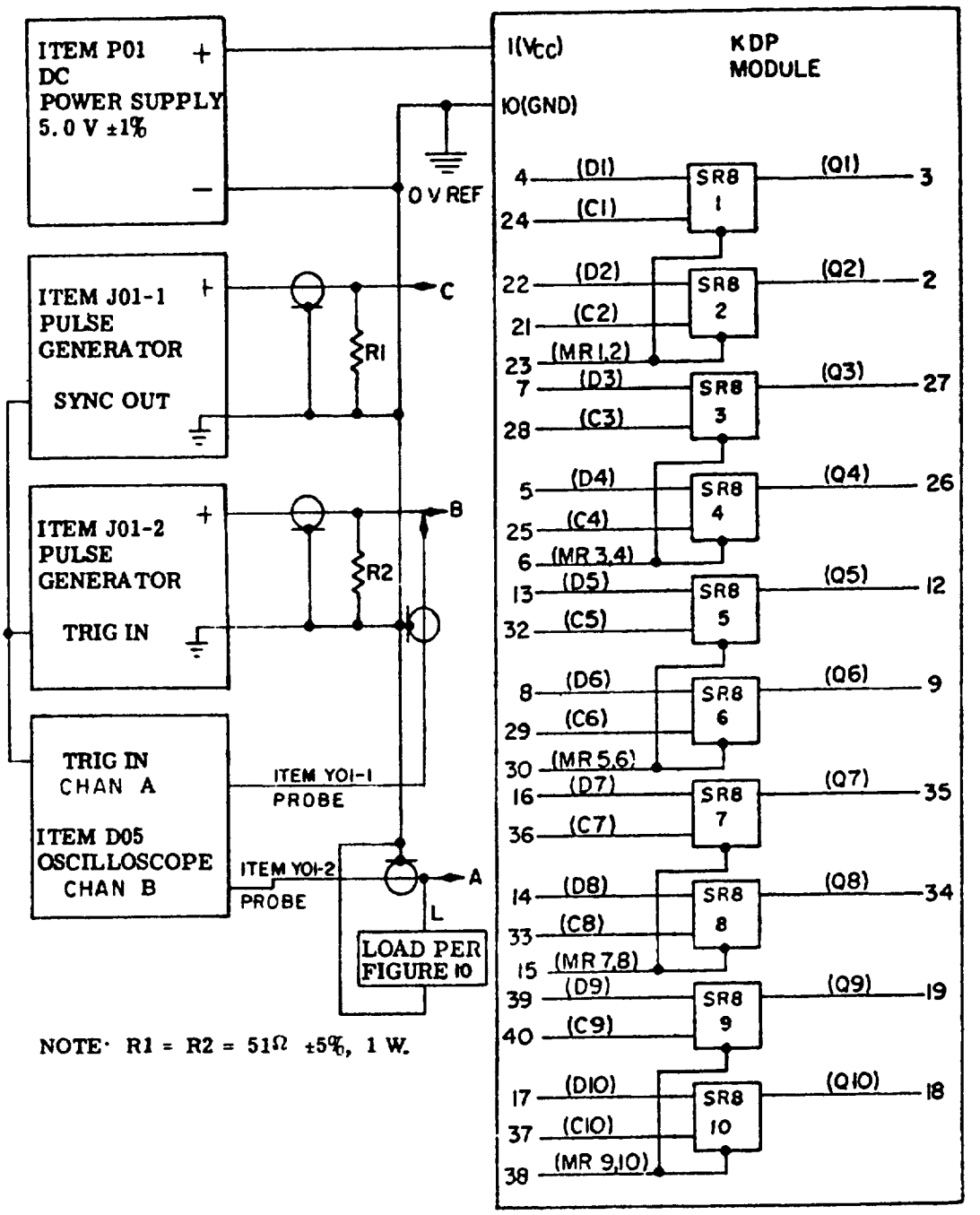
Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
L1	1Q Transition time (t_{TLH})	3	4	24	4	45	ns	15, 16
L2	2Q	2	22	21				
L3	3Q	27	7	28				
L4	4Q	26	5	25				
L5	5Q	12	13	32				
L6	6Q	9	8	29				
L7	7Q	35	16	36				
L8	8Q	34	14	33				
L9	9Q	19	39	40				
L10	10Q	18	17	37				
M1	1Q Transition time (t_{THL})	3	4	24		20		
M2	2Q	2	22	21				
M3	3Q	27	7	28				
M4	4Q	26	5	25				
M5	5Q	12	13	32				
M6	6Q	9	8	29				
M7	7Q	35	16	36				
M8	8Q	34	14	33				
M9	9Q	19	39	40				
M10	10Q	18	17	37				

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* TABLE XI. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
N1	1Q Propagation delay time (t_{PLH})	3	4	24	4	30	ns	15,16
N2	2Q	2	22	21				
N3	3Q	27	7	28				
N4	4Q	26	5	25				
N5	5Q	12	13	32				
N6	6Q	9	8	29				
N7	7Q	35	16	36				
N8	8Q	34	14	33				
N9	9Q	19	39	40				
N10	10Q	16	17	37				
P1	1Q Propagation delay time (t_{PHL})	3	4	24		46		
P2	2Q	2	22	21				
P3	3Q	27	7	28				
P4	4Q	26	5	25				
P5	5Q	12	13	32				
P6	6Q	9	8	29				
P7	7Q	35	16	36				
P8	8Q	34	14	33				
P9	9Q	19	39	40				
P10	10Q	18	17	37				

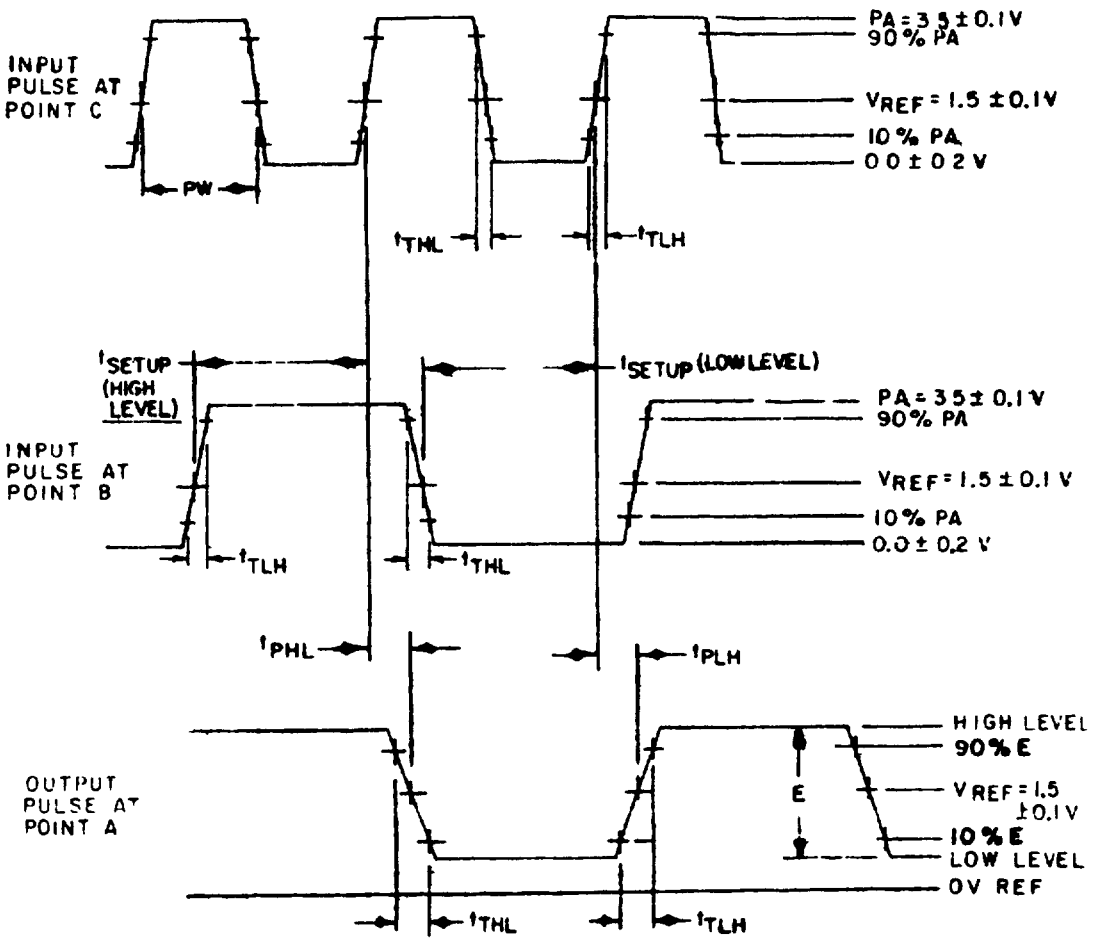
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NOTE: R1 = R2 = 51Ω ±5%, 1 W.

* FIGURE 15. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for table XI.

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NOTES

1. Input pulse parameters (point C):
   PW = 500 ns ±10%.
   PRR = 1 MHz ±10%.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. Input pulse parameters (point B):
   t_{SETUP} (HIGH LEVEL) = t_{SETUP} (LOW LEVEL) = 750 ns ±10%.
   PRR = 500 kHz ±10%.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
3. Output pulse parameters (point A):
   High level ≥ 2.4 V.
   Low level ≤ 0.4 V.
   t_{TLH}, t_{THL}, t_{PLH}, and t_{PHL} as specified in table XI.

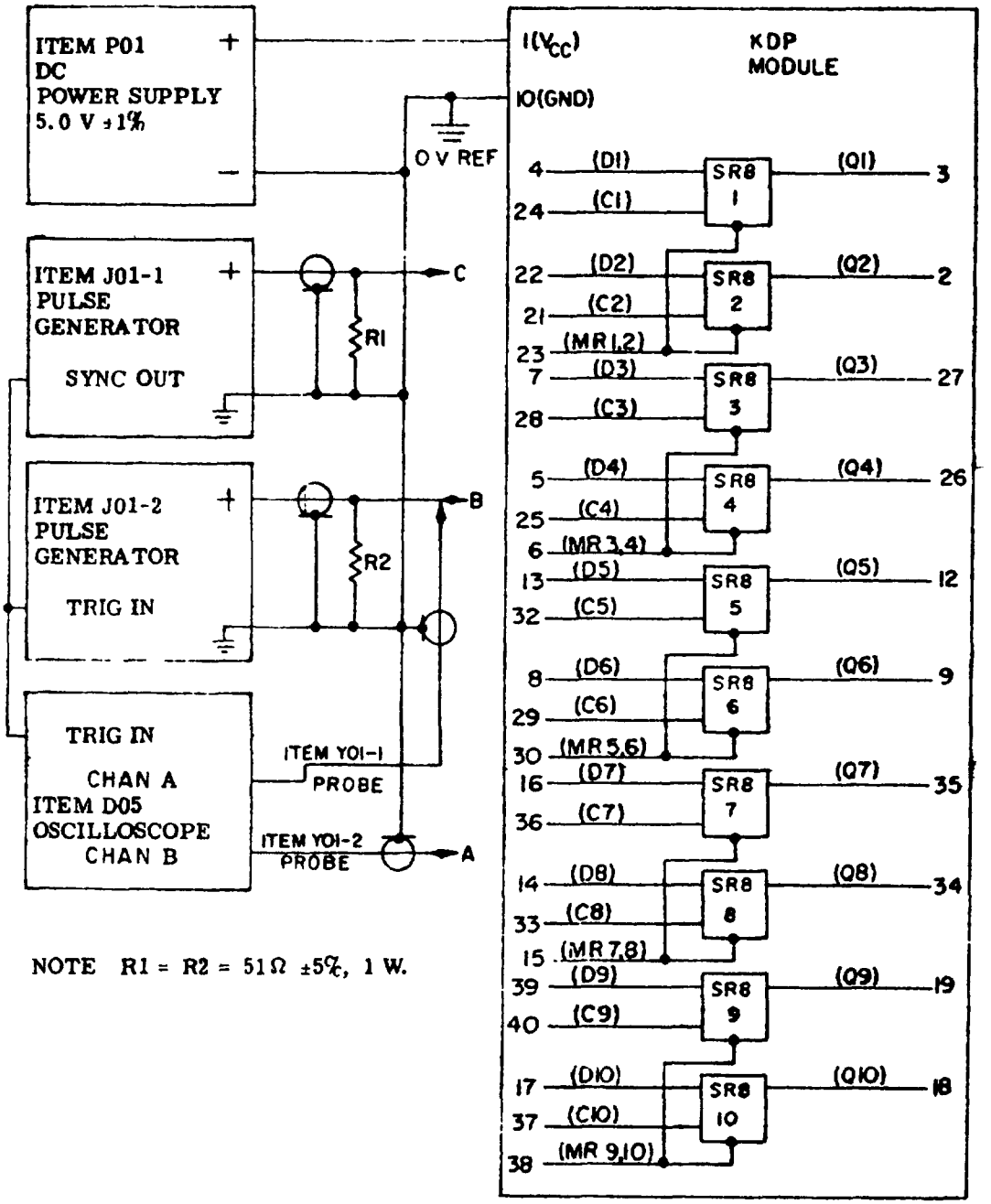
* FIGURE 16. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for table XI.

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TABLE XII. Setup and hold time test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient						Unit	Test figure
		A to pin	B to pin	C to pin	TA		TB		TC			
					Min	Max	Min	Max	Min	Max		
R1	1Q Setup time	3	4	24	487	637	432	590	900	1100	ns	17, 18
R2	2Q	2	22	21								
R3	3Q	27	7	28								
R4	4Q	26	5	25								
R5	5Q	12	13	32								
R6	6Q	9	8	29								
R7	7Q	35	16	36								
R8	8Q	34	14	33								
R9	9Q	19	39	40								
R10	10Q	18	17	37								
S1	1Q Hold time	3	4	24	0	50						17,19
S2	2Q	2	22	21								
S3	3Q	27	7	28								
S4	4Q	26	5	25								
S5	5Q	12	13	32								
S6	6Q	9	8	29								
S7	7Q	35	16	36								
S8	8Q	34	14	33								
S9	9Q	19	39	40								
S10	10Q	18	17	37								

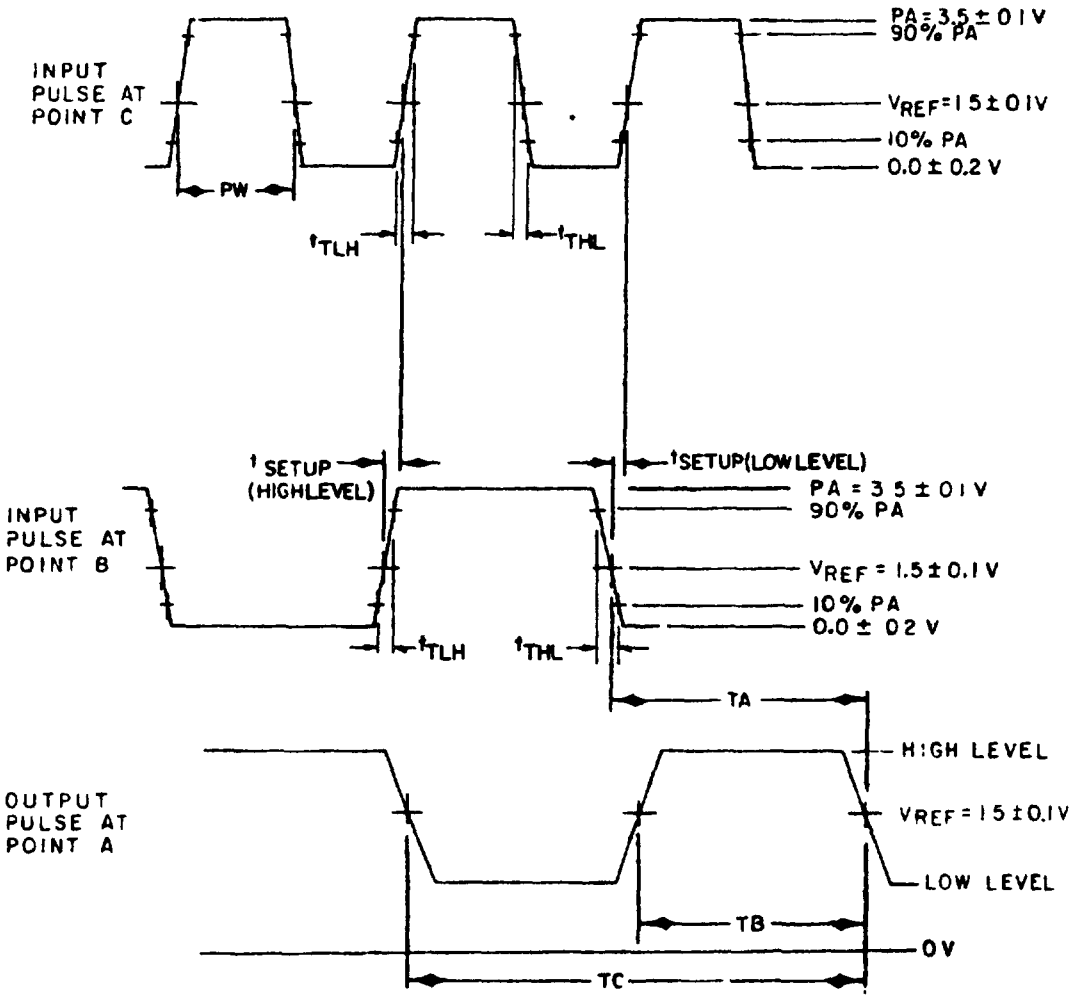
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NOTE R1 = R2 = 51Ω ±5%, 1 W.

* FIGURE 17. Setup time and hold time test circuit for table XII.

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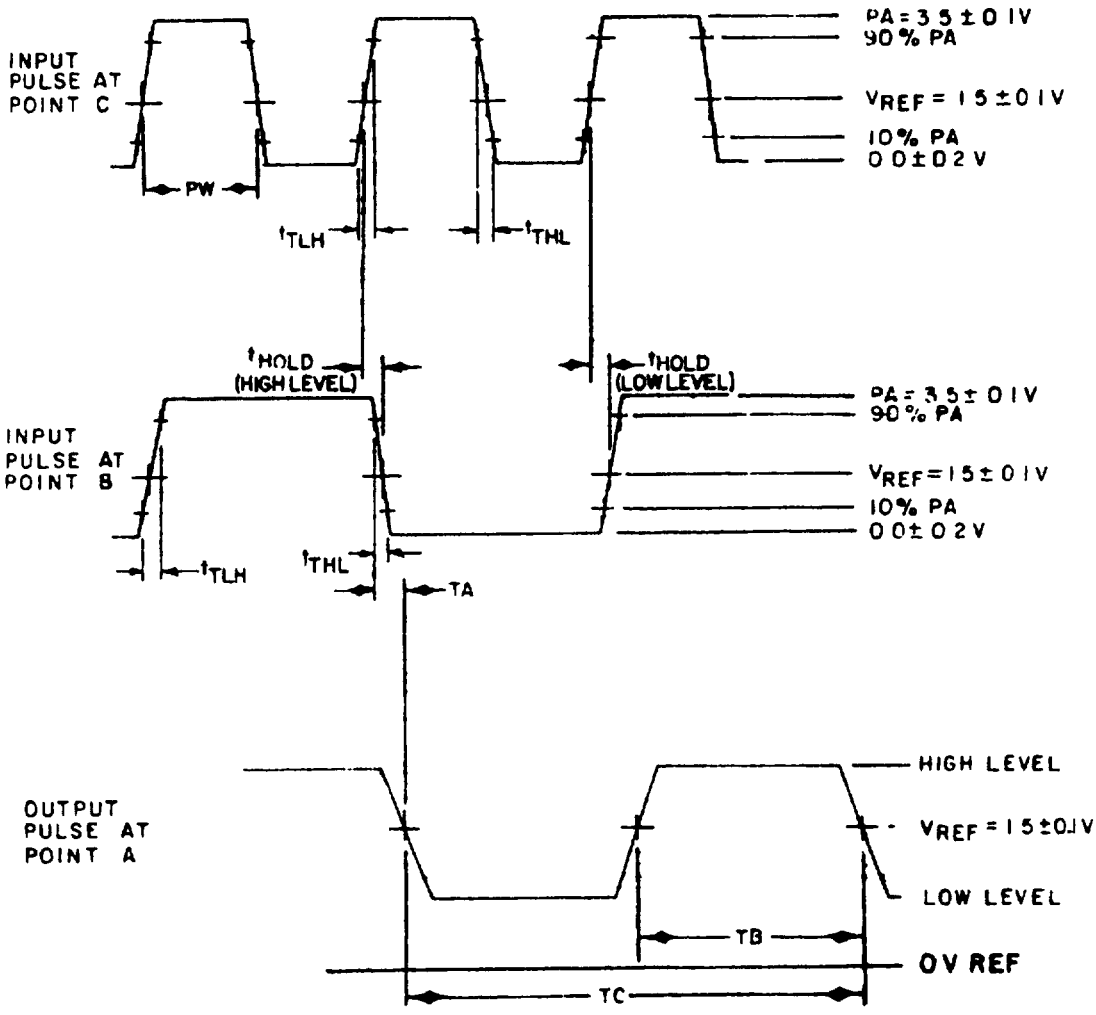


NOTES

1. Input pulse parameters (point C):
   PW = 250 ns ±10%.
   PRR = 2 MHz ±10%.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. Input pulse parameters (point B):
   t_{SETUP}(HIGH LEVEL) = t_{SETUP}(LOW LEVEL) = $35\ ^{+?}_{-0}$ ns.
   PRR = 1 MHz ±10%.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
3. Output pulse parameters (point A)
   High level ≥ 2.4 V.
   Low level ≤ 0.4 V.
   TA, TB, and TC as specified in table XII.

* FIGURE 18. Setup time test circuit waveforms for table XII.

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NOTES:
1. Input pulse parameters (point C)
   PW = 250 ns ±10%.
   PRR = 2 MHz ±10%.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. Input pulse parameters (point B):
   t_{HOLD}(HIGH LEVEL) = t_{HOLD}(LOW LEVEL) = 0 $^{+2}_{-0}$ ns.
   PRR = 1 MHz ±10%.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
3. Output pulse parameters (point A):
   High level ≥ 2.4 V.
   Low level ≤ 0.4 V.
   TA, TB, and TC as specified in table XII.

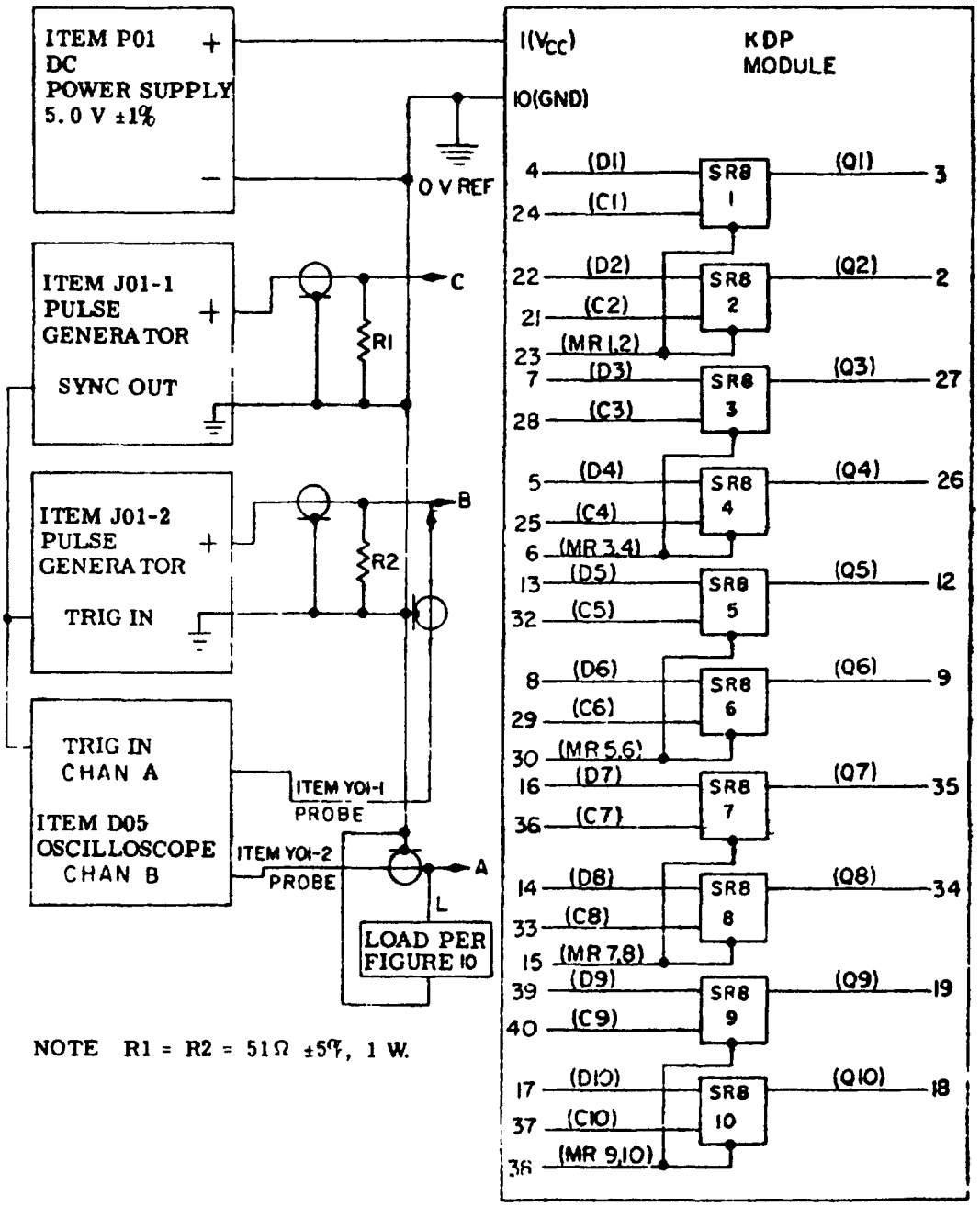
* FIGURE 19. Hold time test circuit waveforms for table XII.

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TABLE XIII. Maximum shift frequency test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient						Unit	Test figure
		A to pin	B to pin	C to pin	TA		TB		TC			
					Min	Max	Min	Max	Min	Max		
T1	1Q Maximum shift frequency	3	4	24	142	205	80	135	180	204	ns	20,21
T2	2Q	2	22	21								
T3	3Q	27	7	28								
T4	4Q	26	5	25								
T5	5Q	12	13	32								
T6	6Q	9	8	29								
T7	7Q	35	16	36								
T8	8Q	34	14	33								
T9	9Q	19	39	40								
T10	10Q	18	17	37								

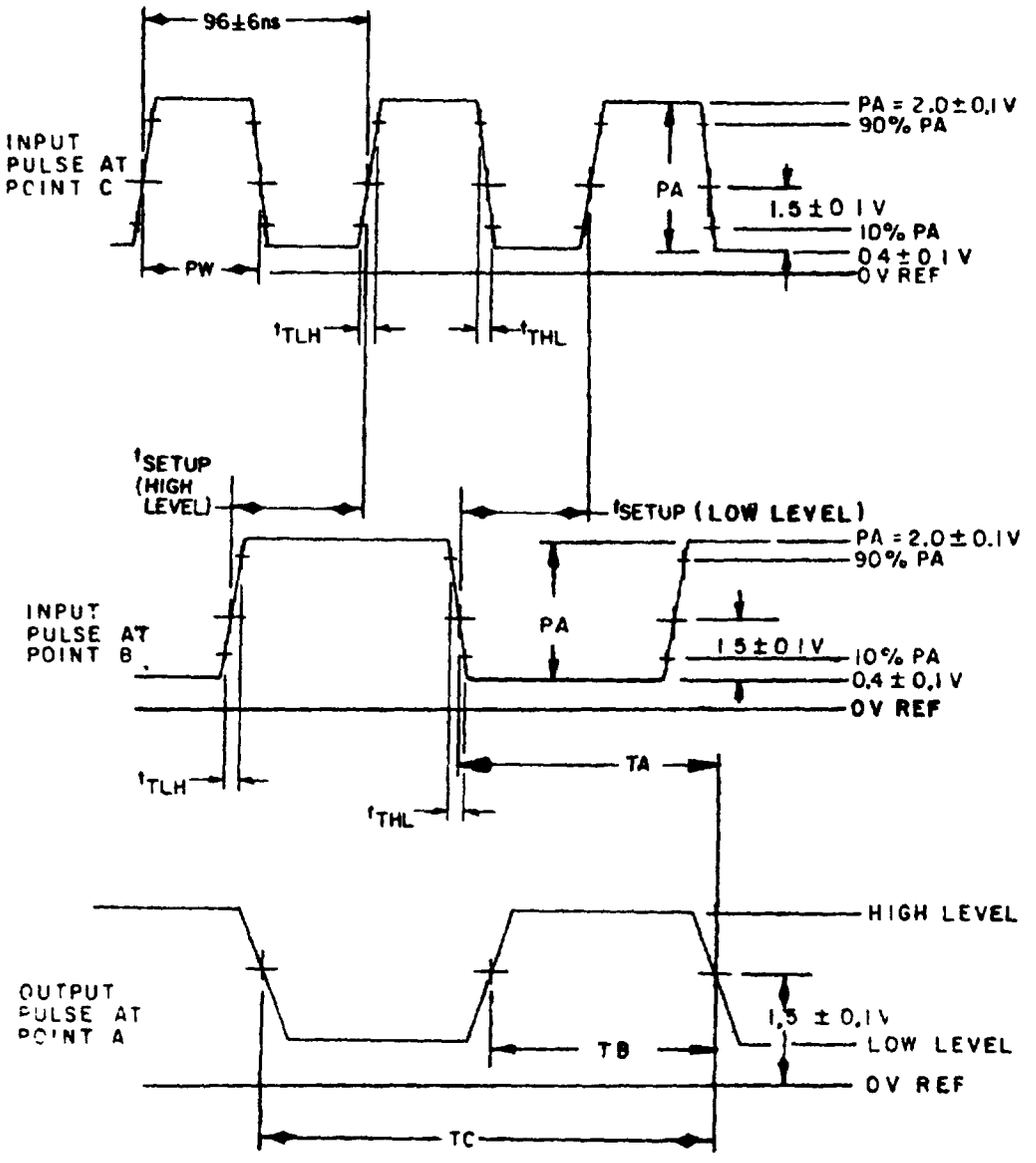
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NOTE R1 = R2 = 51Ω ±5%, 1 W.

* FIGURE 20. Maximum shift frequency test circuit for table XIII

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NOTES

1. Input pulse parameters (point C):
   P_W = 48 ±3 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. Input pulse parameters (point B):
   t_{SETUP} (HIGH LEVEL) = t_{SETUP} (LOW LEVEL) = 50 ns ±5%.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
3. Output pulse parameters (point A):
   High level ≥ 2.4 V.
   Low level ≤ 0.4 V.
   TA, TB, and TC as specified in table XIII.

* FIGURE 21. Maximum shift frequency test circuit waveforms for table XIII.

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*TABLE XIV. Propagation delay time (t_{PHL}) (minimum MR pulsewidth) test.

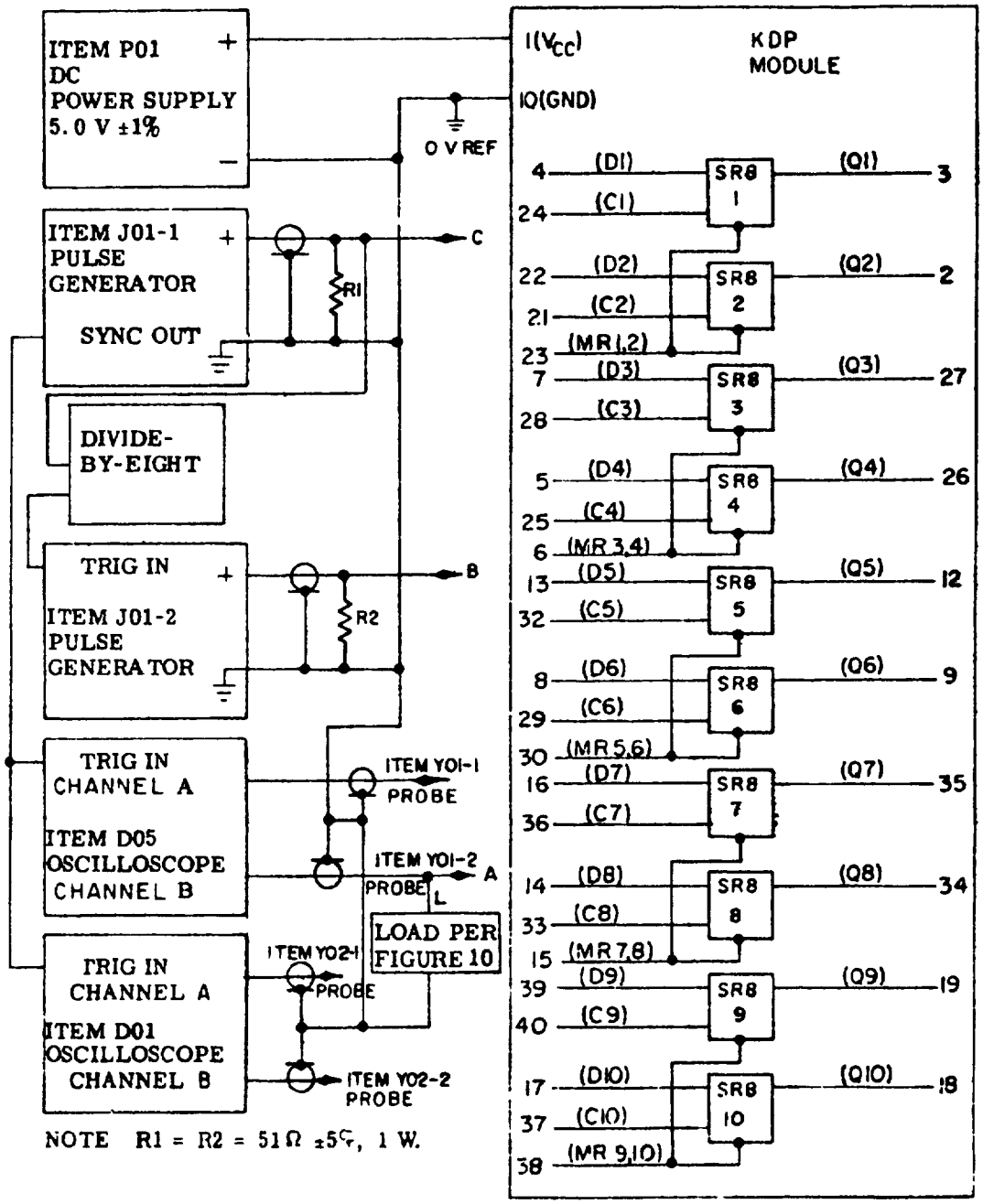
Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J1	1Q Propagation delay time (t_{PHL}), (minimum MR pulse-width)	3	23	24	4	75	ns	22,23
U2	2Q	2	23	21				
J3	3Q	27	6	28				
U4	4Q	26	6	25				
U5	5Q	12	30	32				
U6	6Q	9	30	29				
U7	7Q	35	15	36				
U8	8Q	34	15	33				
U9	9Q	19	38	40				
U10	10Q	18	38	37				

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* TABLE XV. Minimum MR pulsewidth (clock low) test.

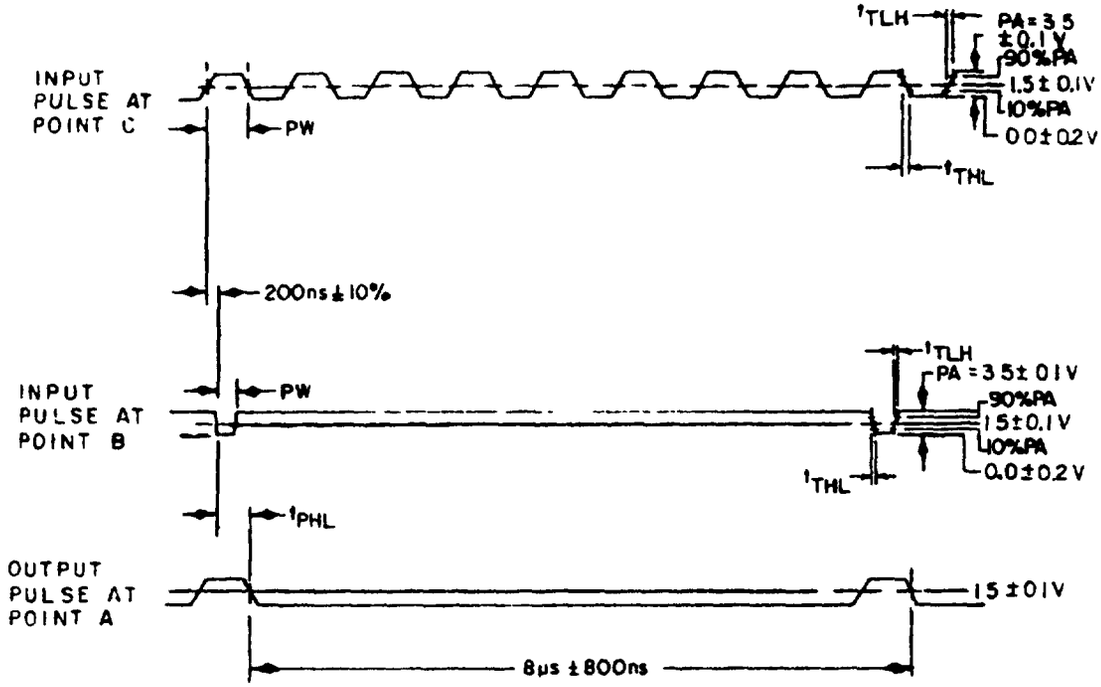
Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
V1	1Q Minimum MR pulsewidth, clock low	3	23	24	7.2	8.8	µs	22,24
V2	2Q	2	23	21				
V3	3Q	27	6	28				
V4	4Q	26	6	25				
V5	5Q	12	30	32				
V6	6Q	9	30	29				
V7	7Q	35	15	36				
V8	8Q	34	15	33				
V9	9Q	19	38	40				
V10	10Q ▼	18	38	37	▼	▼	▼	▼

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* FIGURE 22. Propagation delay time (t_{PHL}) (minimum MR pulsewidth) and minimum MR pulsewidth (clock low) test circuit for tables XIV and XV.

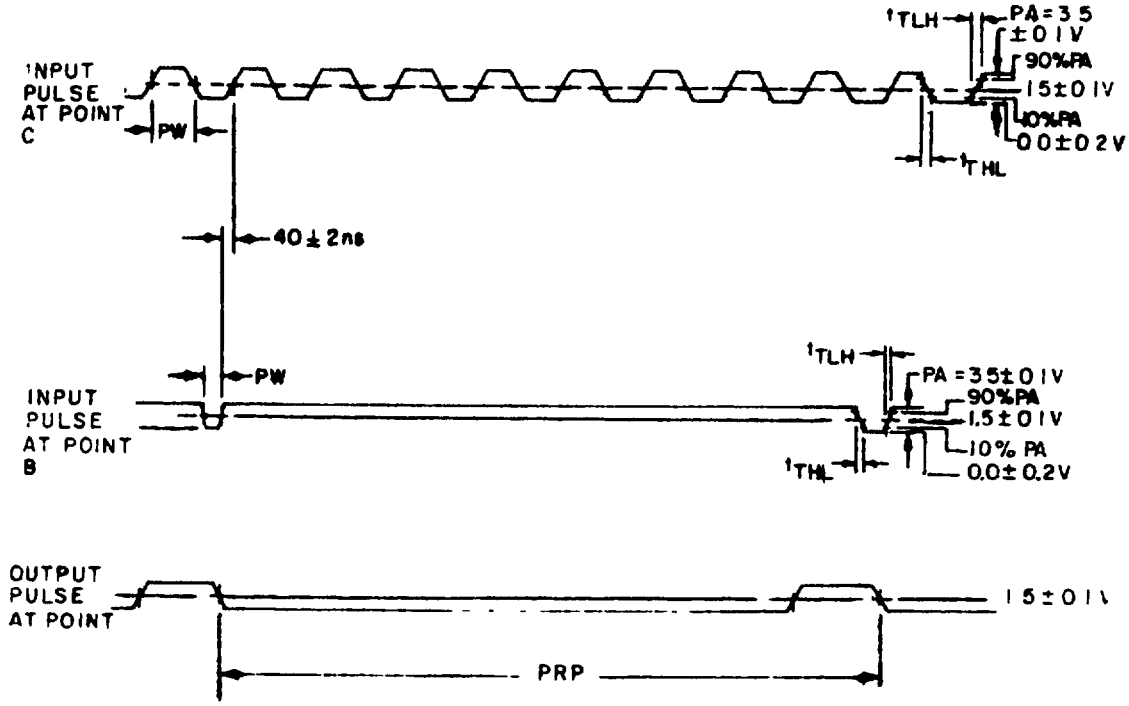
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NOTES.
1. Input pulse parameters (point C):
   PW = 500 ns ±10%.
   PRR = 1 MHz ±10%.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. Input pulse parameters (point B):
   PW = $35\ ^{+4}_{-0}$ ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
3. Output pulse parameters (point A):
   t_{PHL} as specified in table XIV.
   PRP = 8 µs ±800 ns.

* FIGURE 23. Propagation delay time (t_{PHL}) (minimum MR pulse width) test circuit waveforms for table XIV.

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NOTES:
1. Input pulse parameters (point C):
   PW = 500 ns ±10%.
   PRR = 1 MHz ±10%.
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
2. Input pulse parameters (point B):
   PW = $45 \, {}^{+4}_{-0}$ ns.
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
3. Output pulse parameters (point A):
   PRP as specified in table XV.

* FIGURE 24. Minimum MR pulsewidth (clock low) test circuit waveforms for table XV.

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TABLE XVI Output crosstalk (high-level) test.

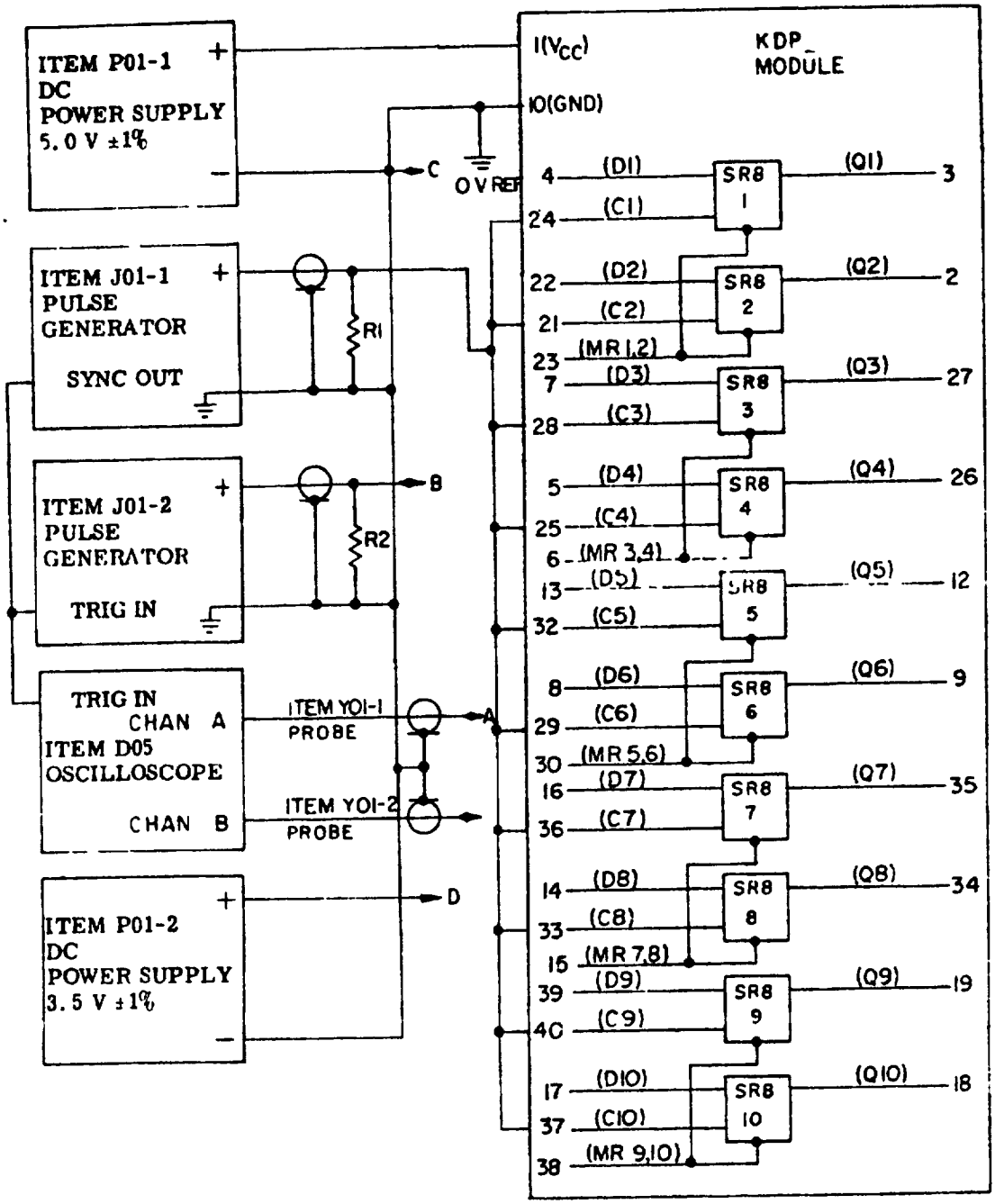
Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
W1	1Q Output crosstalk, high level	3	22,7,5,13, 8,16,14, 39,17	Open	4	0	400	mV	25,26
W2	2Q	2	4,7,5,13, 8,16,14, 39 17		22				
W3	3Q	27	4,22,5,13, 8,16,14, 39,17		7				
W4	4Q	26	4,22,7,13, 8,16,14, 39,17		5				
W5	5Q	12	4,22,7,5, 8,16,14, 39,17		13				
W6	6Q	9	4,22,7,5, 13,16,14, 39,17		8				
W7	7Q	35	4,22,7,5, 13,8,16, 39,17		16				
W8	8Q	34	4,22,7,5, 13,8,16, 39,17		14				
W9	9Q	19	4,22,7,5, 13,8,16, 14,17		39				
W10	10Q	18	4,22,7,5, 13,8,16, 14,39		17				

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TABLE XVII. Output crosstalk (low-level) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
X1	10 Output crosstalk, low level	3	22,7,5,13, 8,16,14, 39,17	4	Open	0	500	mV	25,26
X2	20	2	4,7,5,13, 8,16,14, 39,17	22					
X3	30	27	4,22,5,13, 8,16,14, 39,17	7					
X4	40	26	4,22,7,13, 8,16,14, 39,17	5					
X5	50	12	4,22,7,5, 8,16,14, 39,17	13					
X6	60	9	4,22,7,5, 13,16,14, 39,17	6					
X7	70	35	4,22,7,5, 13,8,16, 39,17	16					
X8	80	34	4,22,7,5, 13,8,16, 39,17	14					
X9	90	19	4,22,7,5, 13,8,16, 14,17	39					
X10	100 ▼	18	4,22,7,5, 13,8,16, 14,39	17	▼	▼	▼	▼	▼

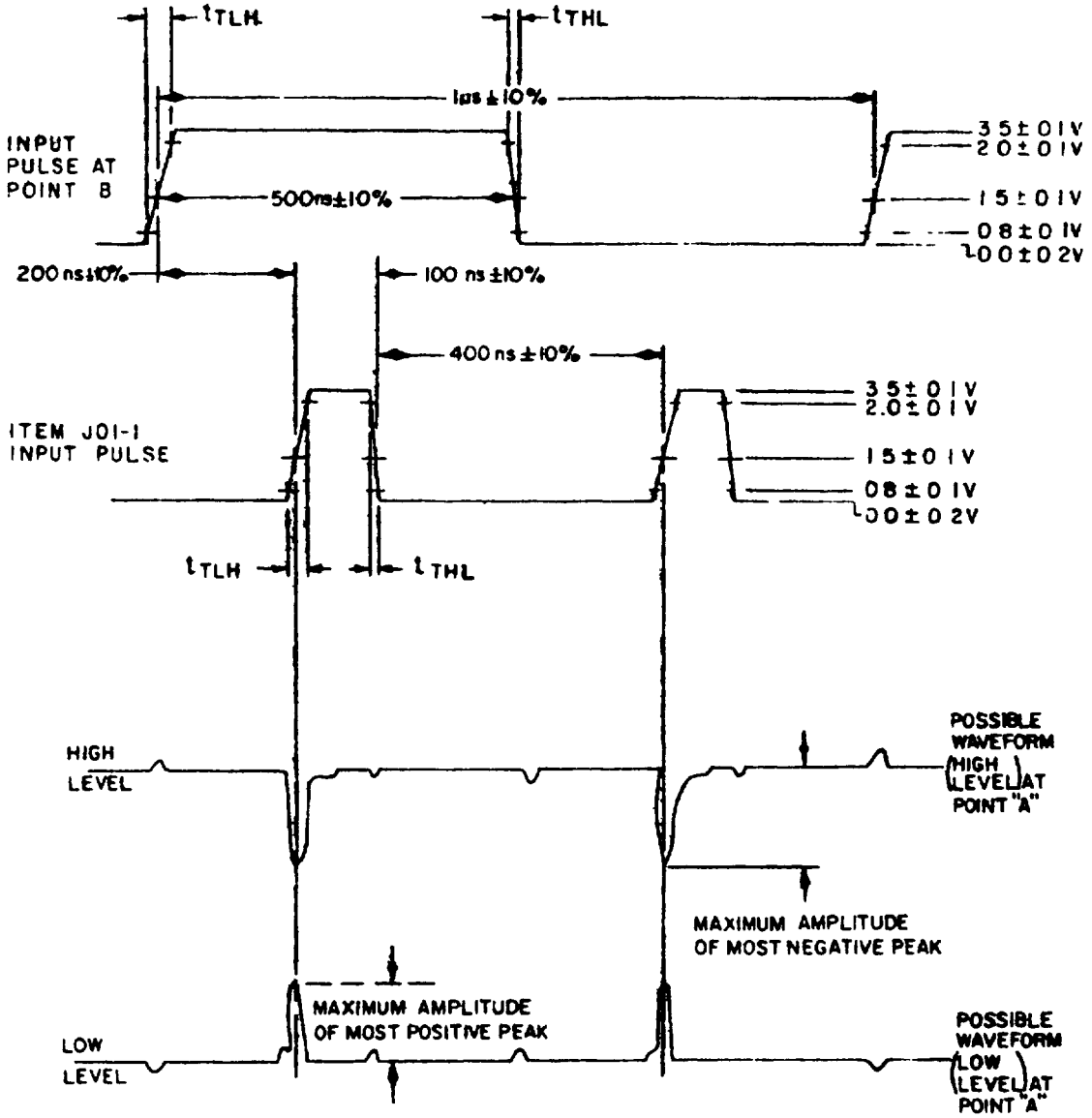
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NOTES
1. R1 - R2 = 51Ω ±5 , 1 W.
2. Load all outputs with load as specified on figure 10.

* FIGURE 25. Output crosstalk (high-level and low-level) test circuit for tables XVI and XVII.

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NOTE $t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns (verified at each input pin).

* FIGURE 26. Output crosstalk (high-level and low-level) test circuit waveforms for tables XVI and XVII.

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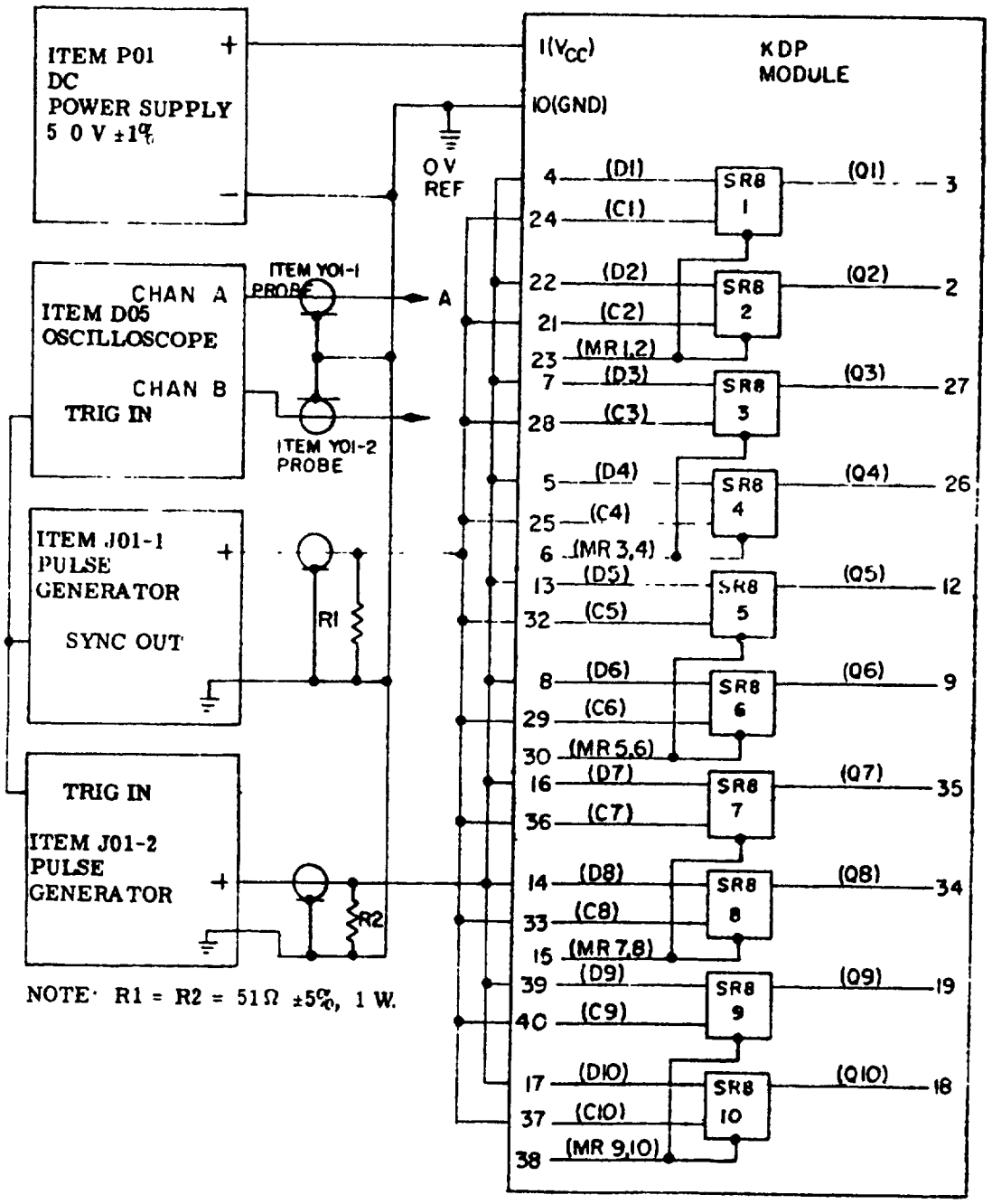
TABLE XVIII. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
Y	Power supply transient amplitude	0	600	mV	27,28

TABLE XIX. Power supply transient rise time test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
Z	Power supply transient rise time	1.0	2.0	ms	29,30

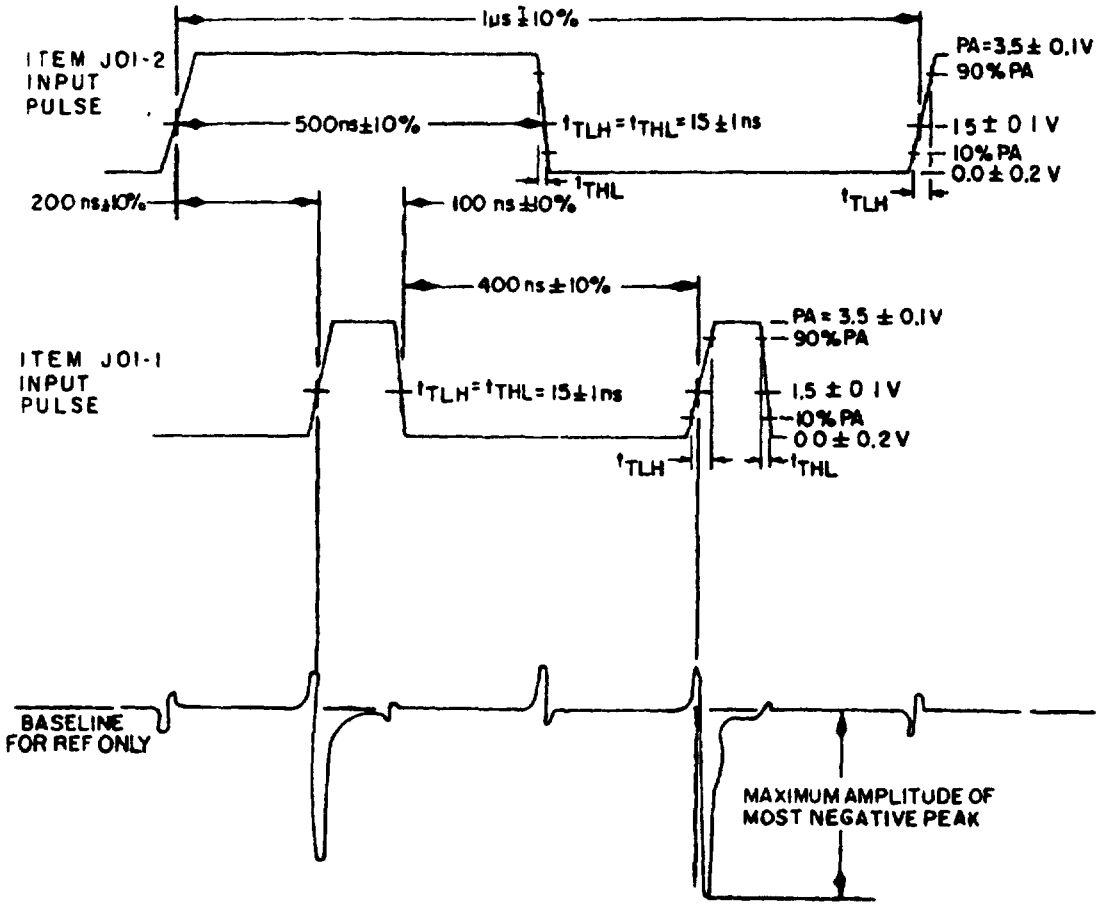
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NOTE: R1 = R2 = 51Ω ±5%, 1 W.

✱ FIGURE 27 Power supply transient amplitude test circuit for table XVIII.

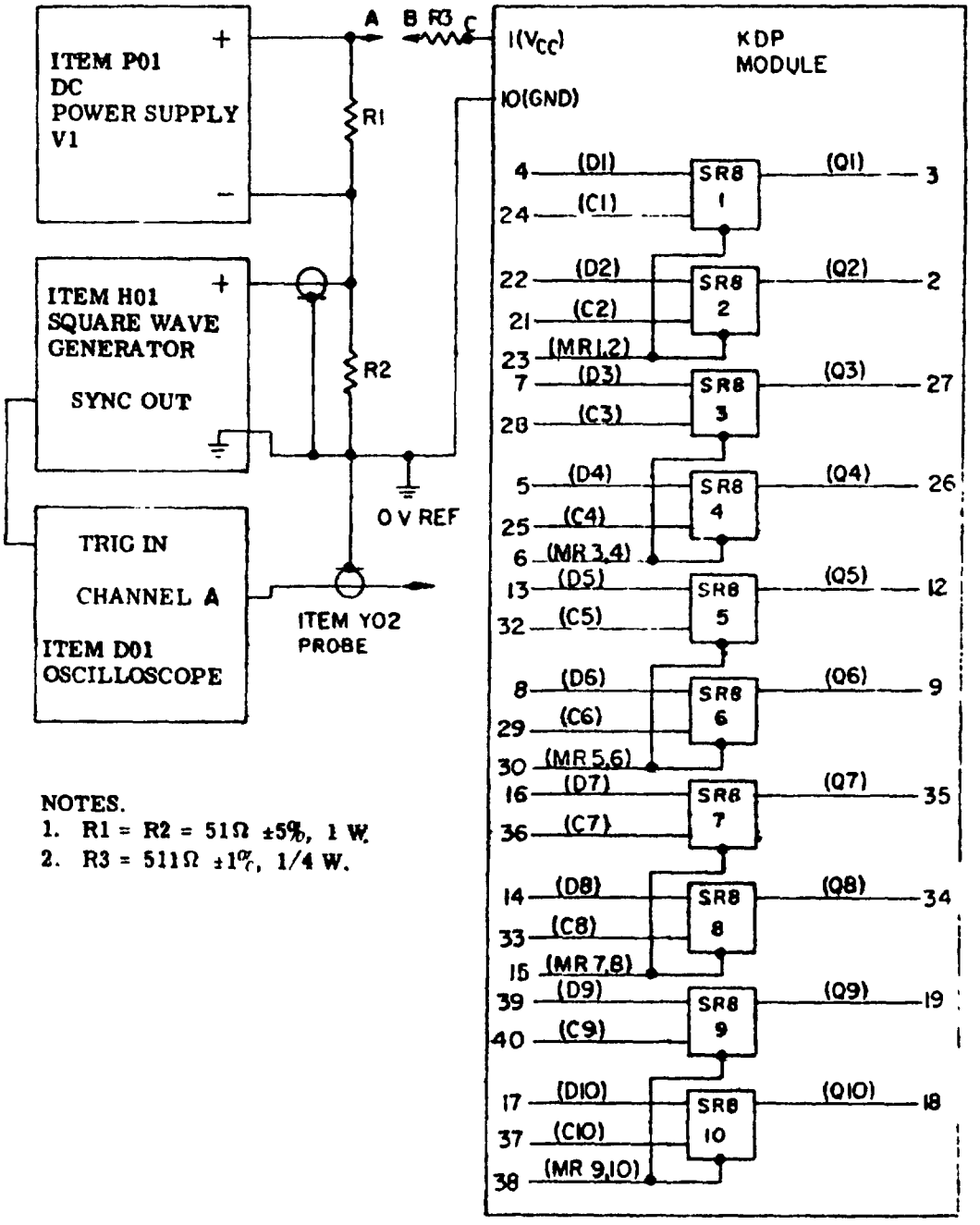
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NOTES:
1. Input pulse parameter:
   $t_{TLH} = t_{THL} = 15 \pm 1$ ns.
2. Connect point A as close as possible to pin 1 on the module. Connect probe ground as close as possible to pin 10 on the module.

* FIGURE 28. Power supply transient amplitude test circuit waveforms for table XVIII.

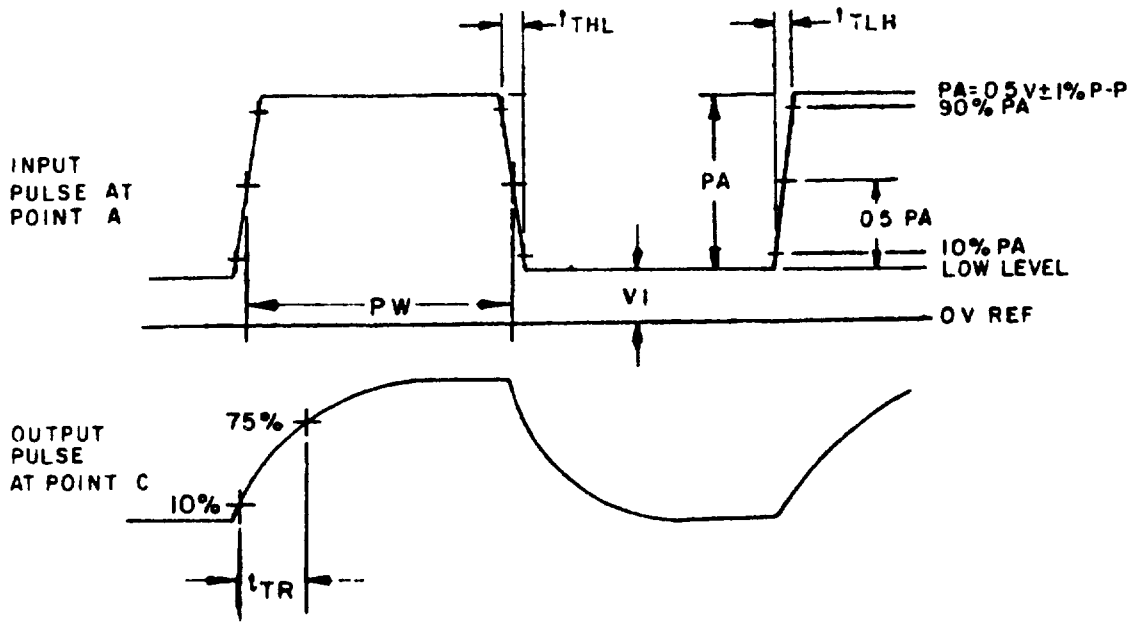
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NOTES.
1. R1 = R2 = 51Ω ±5%, 1 W.
2. R3 = 511Ω ±1%, 1/4 W.

* FIGURE 29. Power supply transient rise time test circuit for table XIX.

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NOTES
1. Input pulse parameters
   PW = 10 ms ±10%.
   PRR = 50 Hz (reference).
   V1 = 0.00 ±0.01 V.
   $t_{TLH} = t_{THL} < 100$ ns (reference).
2. With point A open, adjust input as shown.
3. Connect A to B and recheck the low level (V1) and pulse parameters at point A.
4. Measure transient rise time (t_{TR}) at point C.

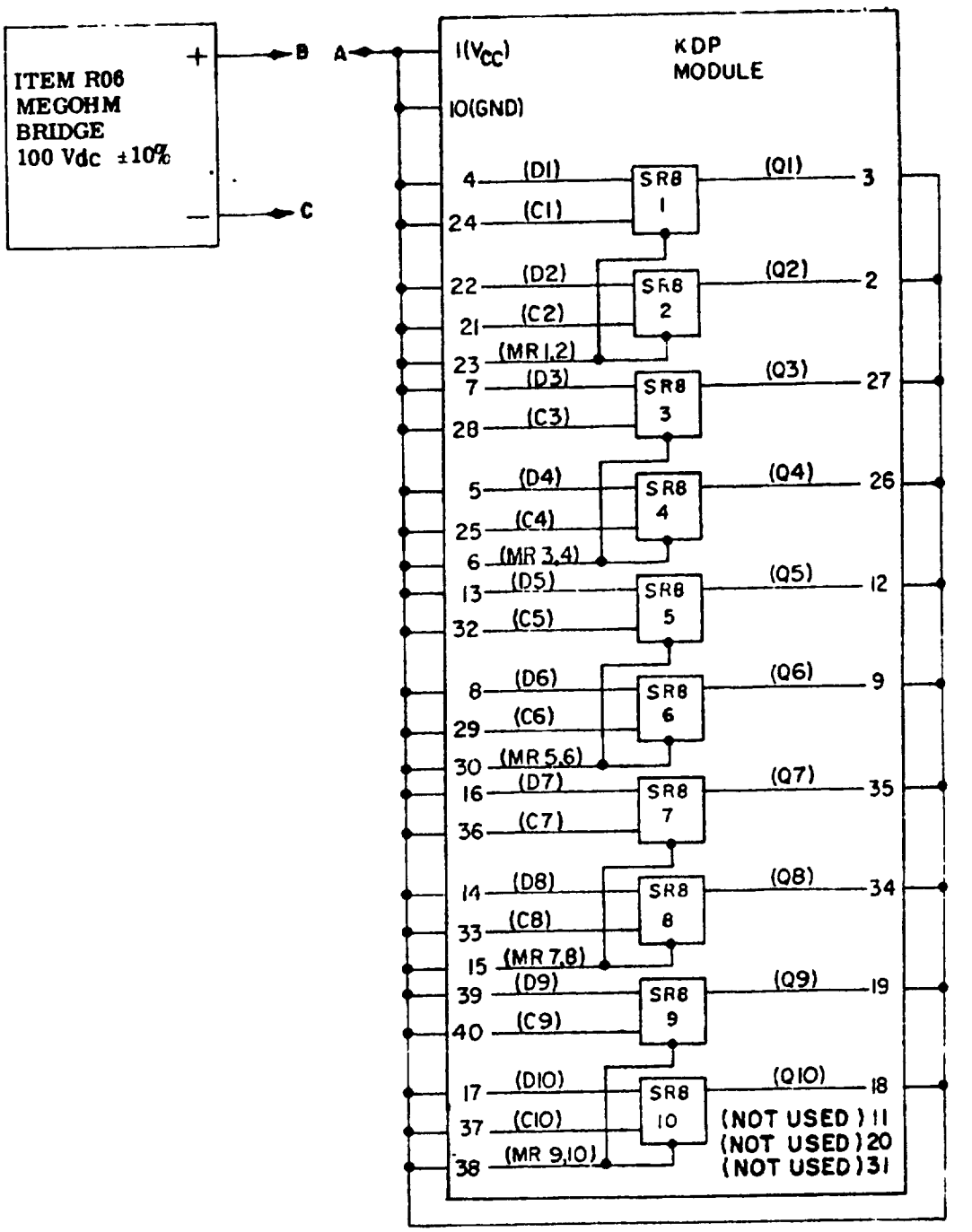
*FIGURE 30. Power supply transient rise time test circuit waveforms for table XIX.

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★ TABLE XX. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		B to pin	C to pin	Min	Max		
AA1	Isolation	A,11,20,31	Frame	10	∞	MΩ	31
AA2	↓	A	11,20,31	↓	↓	↓	↓
AA3		11	20,31				
AA4		20	31				

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WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE For test condition AA1 penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

* FIGURE 31. Isolation test circuit for table XX.

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TABLE XXI. Input capacitance test.

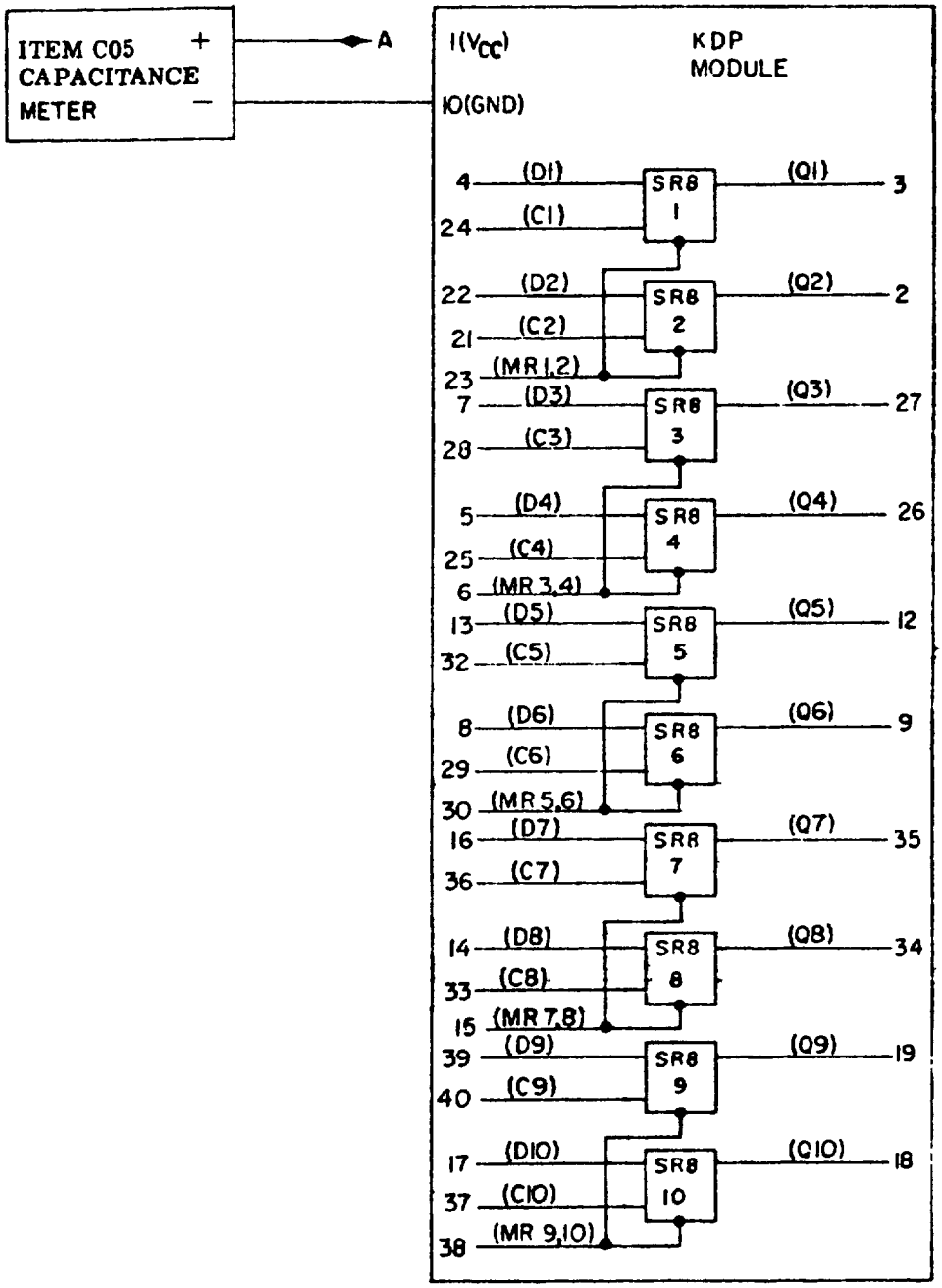
Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
BB1	1D Input capacitance	4	2	15	pF	32
BB2	1C	24				
BB3	2D	22				
BB4	2C	21				
BB5	1, 2MR	23				
BB6	3D	7				
BB7	3C	28				
BB8	4D	5				
BB9	4C	25				
BB10	3, 4MR	6				
BB11	5D	13				
BB12	5C	32				
BB13	6D	8				
BB14	6C	29				
BB15	5, 6MR	30				
BB16	7D	16				
BB17	7C	36				
BB18	8D	14				
BB19	8C	33				
BB20	7, 8MR	15				
BB21	9D	39				
BB22	9C	40				
BB23	10D	17				

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TABLE XXI. Input capacitance test - Continued.

Test condition	Parameter	Connect	Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
B24	10C Input capacitance	37	2	15	pF	32
B25	9, 10MR Input capacitance	38	2	15	pF	32

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NOTES:
1. Null out effect of test leads prior to connecting to module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.

* FIGURE 32. Input capacitance test circuit for table XXI

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